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Generative AI for RAG based AI system for Automated Telecom Customer Support

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DEDICATION

Dedicated to my family and supervisor Dr. Nitin Arvind Shelke,
whose unwavering support, patience, and belief in my abilities
inspired me to pursue and complete this research journey.

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ABSTRACT

The telecommunications industry is experiencing rapid growth in the number of customer interactions, making it increasingly difficult for traditional customer support systems to provide fast, accurate, and personalized assistance. Most existing support solutions rely on rule-based chatbots or keyword-based search systems, which often fail to understand the context of customer queries. As a result, customers may receive inaccurate responses, experience longer waiting times, and require repeated interactions before their issues are resolved. These challenges highlight the need for more intelligent and context-aware customer support systems. This research proposes a Retrieval-Augmented Generation (RAG)-based Artificial Intelligence framework to improve automated customer support in the telecommunications sector. The proposed system combines the language understanding capabilities of Large Language Models (LLMs) with a telecom-specific knowledge base to generate accurate and reliable responses. The knowledge base consists of telecom documents such as FAQs, troubleshooting guides, billing information, service plans, policy documents, and technical manuals. These documents are converted into vector embeddings and stored in a vector database, allowing the system to retrieve the most relevant information before generating a response. By grounding the generated answers in trusted company knowledge, the framework aims to reduce AI hallucinations while improving response quality and consistency. The performance of the proposed framework will be evaluated using telecom customer support datasets and measured through key metrics such as retrieval accuracy, response relevance, precision, recall, F1-score, query resolution time, and customer satisfaction. The research also considers important aspects of responsible AI, including data privacy, security, transparency, and governance, to ensure that the proposed solution can be adopted safely within enterprise environments. The expected findings suggest that integrating Retrieval-Augmented Generation with telecom knowledge sources can significantly improve the quality and efficiency of automated customer support. The proposed framework is expected to provide faster and more accurate responses, reduce customer waiting times, improve first-contact resolution, and enhance the overall customer experience while lowering operational costs. In addition, this research offers a practical and scalable AI solution for the telecommunications industry and identifies future opportunities for extending the framework through multilingual support, Agentic AI, real-time knowledge updates, and proactive customer assistance.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
RAG	Retrieval-Augmented Generation
IOT	Internet of Things
SDN	software-defined networking
NFV	network function virtualisation
NLP	Natural Language Processing
LLM	Large Language Models
GENAI	Genrative Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
SVM	Support Vector Machin
TML	Traditional Machine Learning
SNA	Social Network Analysis
KNN	K-Nearest Neighbour
SVC	Support Vector Classification
RNN	Recurrent Neural Network
CNN	Convolutional Neural Network
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimised BERT Pretraining Approach
DistilBERT	Distilled Bidirectional Encoder Representations from Transformers
LLaMA	Large Language Model Meta AI
MRR	Mean Reciprocal Rank
GPT	Generative Pre-trained Transformer

Chapter 1- INTRODUCTION

The twenty-first century has witnessed rapid advancements in Artificial Intelligence (AI), transforming how organisations deliver customer support and business services. In the telecommunications industry, increasing customer expectations demand faster, more accurate, and context-aware assistance than traditional support systems can provide. This introductory chapter establishes the background and motivation for developing a Retrieval-Augmented Generation (RAG)-based AI framework for automated telecom customer support. It outlines the research problem, objectives, scope, significance, and overall structure of the dissertation, providing a clear foundation for the chapters that follow.

1.1 Background of the Study

The telecommunications industry has experienced rapid transformation through the adoption of broadband, cloud computing, Internet of Things (IoT), software-defined networking (SDN), network function virtualisation (NFV), and 5G technologies. These developments have increased network complexity and generated substantial volumes of information, including billing records, service plans, FAQs, troubleshooting guides, policies, and technical standards. Efficient access to this information is therefore essential for delivering accurate and timely customer support.

Traditional customer-support systems typically use rule-based chatbots, keyword matching, and decision-tree approaches. Although effective for simple and repetitive queries, these systems have limitations in understanding complex, ambiguous, and context-dependent customer requests. They may also struggle to maintain conversational context and retrieve relevant information from large knowledge repositories, potentially resulting in incomplete responses and repeated support interactions. Advances in Natural Language Processing (NLP) and Large Language Models (LLMs) have created new opportunities for intelligent customer support. Models such as GPT, Claude, Gemini, and LLaMA can understand natural language and generate coherent responses. However, standalone LLMs may generate inaccurate, outdated, or unsupported information, commonly described as hallucination. Such limitations are particularly important in telecommunications, where incorrect information regarding billing, roaming, services, or technical issues can negatively affect customer experience. Retrieval-Augmented Generation (RAG) addresses these limitations by combining information retrieval with generative AI. RAG retrieves

relevant information from trusted knowledge sources and provides it as contextual evidence to an LLM before response generation. By integrating embedding models, vector databases, semantic retrieval, and reranking, RAG can improve response relevance and grounding. Therefore, this research proposes a Generative AI and RAG-based framework for automated telecom customer support, evaluated using retrieval quality, answer accuracy, response relevance, grounding, latency, and query-resolution measures. The study also considers responsible AI aspects such as privacy, transparency, security, and reliability.

1.2 Problem Statement

Before proposing a solution, it is important to understand the challenges faced by existing telecom customer support systems. Although Artificial Intelligence and LLM have improved automated customer service, many current solutions still struggle to provide accurate and reliable responses. Traditional chatbots often rely on predefined rules or keyword matching, making it difficult to understand complex customer queries or retrieve the right information from large telecom knowledge bases. Similarly, generative AI models can produce incorrect or misleading answers when they are not connected to updated enterprise documents. As telecom services continue to grow in complexity, customers expect quick, personalised, and context-aware support. This research addresses these challenges by developing a (RAG)-based AI framework that combines trusted telecom knowledge with the reasoning capabilities of Large Language Models to generate accurate, reliable, and context-aware customer support responses while reducing AI hallucinations and improving overall service efficiency.

1.3 Aim and Objectives

This section defines the overall aim of the research and the specific objectives that guide the study. Existing telecom customer support systems often rely on rule-based chatbots or standalone (LLMs), which may generate inaccurate responses, lack domain-specific knowledge, and produce hallucinated information. To address these challenges, this research proposes a (RAG)-based framework that combines semantic retrieval with the reasoning capabilities of LLMs to deliver accurate, context-aware, and reliable responses for telecom customer support.

Aim

To design, develop, and evaluate a Retrieval-Augmented Generation (RAG)-based Generative

Artificial Intelligence framework for automated telecom customer support that integrates Large Language Models with telecom-specific knowledge retrieval to provide accurate, context-aware, and reliable responses while reducing hallucinations and improving customer service performance.

Objectives

1. Data Collection: Collect TelecomRAG datasets, 3GPP standards, telecom FAQs, troubleshooting guides, and customer support documents from public academic and industry sources.

2. Data Pre-processing: Clean, normalize, chunk documents, extract metadata, and generate embeddings to create a structured telecom knowledge base.

3. Vector Database Deployment: Store document embeddings in FAISS or ChromaDB for semantic indexing, efficient retrieval, and telecom knowledge management.

4. **RAG Model Design and Development:** Integrate LLMs with RAG, semantic retrieval, and prompt engineering to generate accurate, context-aware telecom customer support responses.

5. Model Evaluation and Performance Analysis: Evaluate using Answer Accuracy, Retrieval Precision, Recall@K, MRR, Response Relevance, Hallucination Rate, and Query Resolution Time.

6. Ethical Considerations and Final Analysis: Assess transparency, fairness, privacy, and responsible AI while benchmarking against rule-based chatbots and standalone LLM systems.

Table 1.1 Summary of Research Objectives

Objective Category	Description	Specific Tasks / Actions
Data Preparation	Collect and preprocess telecom knowledge sources for RAG-based customer support.	• Collect TelecomRAG datasets • Collect 3GPP standards, FAQs, troubleshooting guides and billing documents • Clean, normalize and chunk documents • Generate text embeddings
Data Pre-processing	Prepare telecom knowledge for semantic retrieval and embedding generation.	• Clean and normalize text • Remove duplicates • Chunk documents • Extract metadata • Generate vector embeddings
Vector Database Deployment	Build a semantic knowledge repository for efficient document retrieval.	• Store embeddings in FAISS or ChromaDB • Create semantic index • Enable fast document retrieval

RAG Model Design and Development	Develop an LLM-powered RAG framework for telecom customer support.	• Integrate LLM with RAG • Implement semantic retrieval • Apply prompt engineering • Generate context-aware responses.
Model Evaluation and Performance Analysis	Evaluate retrieval quality, response accuracy, and overall system performance.	• Measure Answer Accuracy • Evaluate Retrieval Precision, Recall@K and MRR • Measure Response Relevance • Assess Hallucination Rate and Query Resolution Time
Ethical Considerations and Final Analysis	Ensure responsible AI practices and compare the proposed framework with existing approaches.	• Evaluate transparency and fairness • Protect privacy • Reduce hallucinations • Compare with rule-based and standalone LLM chatbots

1.4 Research Questions

Having defined the objectives, this section narrows them into focused research questions that guide experimentation and evaluation. Each question links directly to a methodological component later discussed in Chapters 3–5.

RQ1: In what ways can Large Language Models be combined with Retrieval-Augmented Generation and telecom knowledge repositories to provide accurate and context-aware responses for customer support?

RQ2: How does the proposed RAG-enabled support framework perform compared with conventional rule-based chatbot systems when evaluated using response accuracy, retrieval speed, issue resolution, and customer satisfaction?

RQ3: Which retrieval strategy—including hybrid retrieval, dense vector search, knowledge graph integration, or reranking—provides the best balance between retrieval accuracy, hallucination reduction, and inference latency within telecom support environments?

RQ4: What ethical, privacy, security, and governance factors should be addressed before deploying GENAI-powered decision support systems in telecommunications?

1.5 Scope of the Study

To maintain focus, this section defines the boundaries of the research by specifying its inclusions and exclusions. By clearly identifying the study scope, the research maintains methodological

consistency and avoids extending beyond its intended objectives. The investigation is limited to the design, development, and evaluation of a Retrieval-Augmented Generation (RAG)-based Generative Artificial Intelligence framework for automated telecom customer support. The proposed system integrates Large Language Models (LLMs) with telecom-specific knowledge retrieval using publicly available datasets, including TelecomRAG, 3GPP standards, telecom FAQs, troubleshooting manuals, billing documents, and customer support knowledge bases obtained from academic and open-source repositories. The study focuses on semantic retrieval, vector databases, and context-aware response generation to improve factual accuracy and reduce AI hallucinations. Applications such as network fault prediction, customer churn prediction, fraud detection, predictive maintenance, network optimisation, and telecom resource management are excluded from the scope. Likewise, commercial deployment, live CRM integration, billing transactions, voice assistants, multilingual speech processing, and proprietary enterprise data are not considered. All experiments are conducted using publicly available datasets while adhering to responsible AI principles, including transparency, fairness, privacy, and explainability.

1.6 Significance of the Study

Following the scope, this section explains the academic and practical importance of the research. By integrating Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs), the study demonstrates how Generative AI can provide accurate, context-aware, and knowledge-grounded responses while reducing AI hallucinations in telecom customer support. From an academic perspective, it contributes to the growing body of research on RAG, semantic retrieval, and enterprise conversational AI within the telecommunications domain. Practically, it proposes a reproducible framework that telecom service providers can adapt to improve customer support efficiency, response quality, and user satisfaction. In addition, the research promotes the responsible adoption of Generative AI by emphasizing transparency, fairness, privacy, and explainability, thereby supporting the development of trustworthy and scalable AI-powered customer support systems.

1.7 Structure of the Thesis

To conclude the introductory chapter, the overall organization of the thesis is outlined to help readers navigate subsequent sections logically.

- **Chapter 1** Introduces the research background, motivation, objectives, and overall scope.
- **Chapter 2** Presents a detailed literature review covering text-based, multimodal, cross-lingual, and explainable-AI approaches while identifying existing research gaps.
- **Chapter 3** Describes the research methodology, including dataset acquisition, preprocessing, model architecture, fusion mechanisms, evaluation metrics, and explainability tools.
- **Chapter 4** Reports exploratory and descriptive statistics detailed analysis of the dataset used to develop and evaluate the proposed Generative AI for RAG-Based AI System for Automated Telecom Customer Support.
- **Chapter 5** Presents experimental results, comparative analyses, explainability visualizations, and ablation studies.
- **Chapter 6** Summarizes conclusions, key contributions, recommendations, and future research directions.

CHAPTER 2 – LITERATURE REVIEW

The literature review explores how automated telecom customer support has evolved from traditional rule-based systems to advanced Retrieval-Augmented Generation (RAG) and Large Language Model (LLM)-based intelligent support frameworks. This chapter establishes the conceptual foundation for the proposed RAG-based Generative Artificial Intelligence framework by reviewing the evolution of conversational AI, semantic retrieval, vector databases, and telecom-specific knowledge retrieval. It further examines recent developments in enterprise Generative AI, Retrieval-Augmented Generation, responsible AI, and intelligent customer support while identifying existing research gaps that motivate the proposed framework for automated telecom customer support.

2.1 Introduction

The rapid advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP) has transformed automated customer support in telecommunications. Traditional rule-based and keyword-based systems often struggle with complex queries and large knowledge bases (Kim and Lee, 2010; Huang et al., 2015). Recent developments in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) enable more context-aware and knowledge-grounded responses by combining semantic retrieval with trusted information (Lewis et al., 2020; Analytics and Sukhwal, 2024). Telecom-focused approaches, including TelecomRAG (Yilma et al., 2025) and Telco-oRAG (Bornea et al., 2025), further demonstrate the potential of RAG in telecom support. This chapter reviews conversational AI, semantic retrieval, vector databases, LLMs, and RAG, while identifying research gaps motivating the proposed framework.

2.2 Systematic Literature Search Strategy

A structured search was conducted using:

Databases: Google Scholar, IEEE Xplore, SpringerLink, and arXiv.

Keywords:Rag for telecom support, GENAI in telecommunications, large language models, Sematic Search, Enterprise Knowledge Retrieval, Telco-oRAG,Vector Databases.

Inclusion criteria:

- Peer-reviewed journal articles and conference papers (2010–2025)
- Studies focusing on AI-based telecom customer support

- Research on Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), semantic retrieval, and vector databases
- Studies reporting implementation details, datasets, evaluation metrics, and experimental results
- Research published in reputable journals, conferences, and recognized preprint repositories

Exclusion criteria:

- Non-academic Souces,Articles.
- Studies lacking reproducibility
- Research unrelated to telecommunications, conversational AI, or knowledge-grounded response generation

After screening , **30 key relevent papers** were selected covering: Machine learning, deep learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG).

2.3 Machine Learning and Deep Learning Based Approaches

2.3.1 Evolution of AI in Telecom Customer Support

The field of automated telecom customer support has undergone a significant transformation over the past two decades, driven by rapid advances in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP). Initially, telecom customer support primarily relied on manual service representatives, rule-based expert systems, and keyword-driven chatbots that responded to predefined customer requests using static decision rules. Early research focused on analysing structured customer information such as billing records, service usage, customer demographics, and network logs to improve customer retention and operational efficiency. Studies by Kim and Lee (2010) examined customer loyalty through statistical analysis, while Jadhav et al. (2011) applied traditional data mining techniques for telecom customer churn prediction using classification and clustering algorithms. As telecom datasets expanded, conventional machine learning algorithms including Decision Trees, Logistic Regression, Support Vector Machines (SVM), Random Forest, and Navie Bayes became widely adopted for predictive analytics and customer behaviour modelling (Sabbeh et al., 2018; Ullah et al., 2019). Although these approaches achieved encouraging results for customer retention and churn prediction, they remained dependent on structured datasets and handcrafted features, limiting their ability to understand natural language queries or provide intelligent customer assistance.

With increasing computational capability and the availability of large-scale telecom datasets, deep learning approaches gradually replaced conventional machine learning models for several telecom applications. Neural network architectures enabled automatic feature extraction and improved predictive performance without extensive manual feature engineering. Amatare and Ojo (2020) and Fujo et al. (2022) demonstrated that deep learning models outperformed traditional machine learning algorithms for telecom customer churn prediction, while Alkhayrat et al. (2020) applied Autoencoder Neural Networks for customer segmentation and dimensionality reduction. At the same time, Patel and Trivedi (2020) integrated Artificial Intelligence, Machine Learning, Natural Language Processing, and AI-powered chatbots to improve customer engagement and support services. Despite these advances, most deep learning and chatbot systems continued to rely on static knowledge bases and predefined training data, limiting their ability to retrieve updated enterprise knowledge or provide context-aware responses for complex telecom queries. More recently, the emergence of Generative Artificial Intelligence, Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG) has fundamentally changed the direction of intelligent telecom customer support. Instead of generating responses solely from model parameters, RAG combines semantic retrieval with external telecom knowledge repositories, enabling AI systems to access current technical documentation, troubleshooting manuals, billing information, and 3GPP standards before generating responses. Studies by Yilma et al. (2025), Bornea et al. (2025), Analytics and Sukhwal (2024), and Yuan et al. (2025) demonstrate that integrating vector databases, semantic retrieval, and Large Language Models significantly improves response relevance, factual accuracy, and knowledge grounding while reducing AI hallucinations. This evolution reflects a broader shift from predictive machine learning models towards intelligent, context-aware, and knowledge-grounded conversational AI systems capable of supporting modern enterprise telecommunications. Figure 2.1 illustrates the progression of automated telecom customer support from traditional machine learning approaches to Retrieval-Augmented Generation-based Generative AI frameworks.



Figure 2.1 – Timeline of RAG based AI system for Automated Telecom Customer Support

2.3.2 Traditional Machine Learning Approaches

Traditional Machine Learning (TML) approaches established the foundation for applying Artificial Intelligence in the telecommunications industry, particularly for customer analytics and decision support. Before the emergence of deep learning and Generative AI, telecom service providers primarily relied on supervised learning algorithms to analyse structured customer data and improve business performance. These approaches focused on applications such as customer churn prediction, customer retention, customer segmentation, service quality prediction, and customer behaviour analysis using historical customer records, billing information, service usage patterns, and network performance data. Jadhav (2011) demonstrated that data mining techniques combined with effective data preprocessing and feature selection significantly improved customer churn prediction using classification algorithms. As telecom datasets increased in size and complexity, conventional machine learning algorithms including Logistic Regression, Decision Trees, Support Vector Machines (SVM), Naïve Bayes, Random Forest, and XGBoost became widely adopted because of their simplicity, interpretability, and predictive capability.

Subsequent research expanded the application of machine learning techniques to large-scale telecom datasets. Sabbeh (2018) compared several classification algorithms for customer retention and reported that ensemble methods, particularly Random Forest and AdaBoost, consistently achieved higher prediction accuracy than individual classifiers. Similarly, Ullah et al. (2019) evaluated Random Forest, J48 Decision Tree, Logistic Regression, Naïve Bayes, and K-Nearest Neighbour using real-world telecom customer datasets, concluding that ensemble learning methods produced superior predictive performance. Ahmad et al. (2019) further enhanced telecom analytics by integrating Apache Spark with machine learning algorithms such as XGBoost and Social Network Analysis (SNA), demonstrating improved scalability and accuracy for customer churn prediction. Despite their effectiveness, these traditional ML approaches relied heavily on structured datasets and manual feature engineering, limiting their ability to understand natural language, retrieve enterprise knowledge, or support conversational customer interactions. Consequently, the limitations of conventional machine learning motivated the development of deep learning, transformer-based models, and Retrieval-Augmented Generation (RAG) frameworks capable of delivering intelligent, context-aware, and knowledge-grounded telecom

customer support. Figure 2.2 illustrates the taxonomy of traditional machine learning approaches in telecommunications, tracing the evolution from statistical and classical supervised learning techniques to ensemble-based and scalable machine learning approaches used for telecom analytics and customer-related applications.

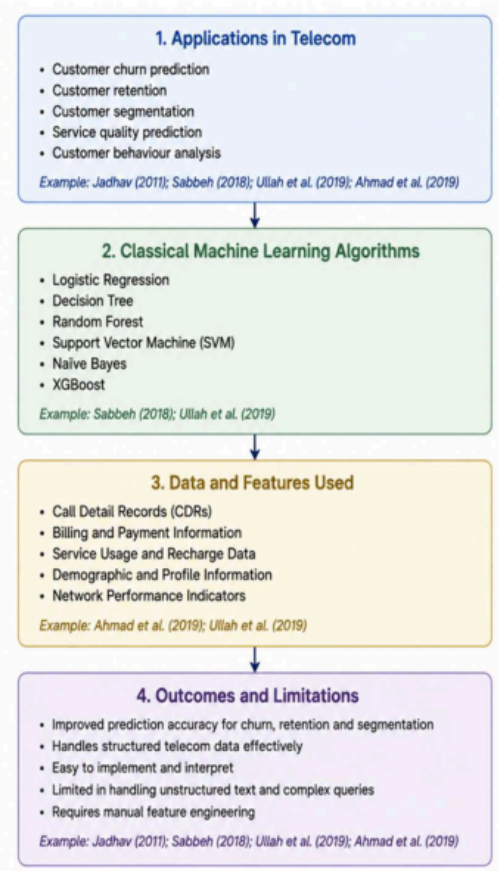


Figure 2.2 – Taxonomy of Traditional Machine Learning Approaches

2.3.3 Natural Language Processing-Based Approaches

Natural Language Processing (NLP) has played an important role in the development of automated telecom customer support by enabling Artificial Intelligence systems to analyse, interpret, and respond to customer-generated text. Early customer-support systems primarily relied on keyword matching, predefined rules, and decision-tree structures, which were effective for simple and repetitive queries but struggled with ambiguity, conversational context, and variations in customer language. As the volume of unstructured customer interactions increased, NLP-based approaches enabled support systems to process customer messages, identify relevant information, and support

automated classification and response generation. The limitations of rule-based and keyword-driven systems are particularly significant in telecommunications, where customer queries frequently contain technical terminology, abbreviated expressions, service-specific language, and context-dependent descriptions of problems.

Patel and Trivedi (2020), in "Leveraging predictive modeling, machine learning personalization, NLP customer support, and AI chatbots to increase customer loyalty," investigated the integration of Artificial Intelligence, Machine Learning, NLP, and AI-powered chatbots for customer support and customer engagement. Their approach incorporated Logistic Regression, K-Nearest Neighbour (KNN), Support Vector Classification (SVC), Decision Tree, Random Forest, AdaBoost, Extra Trees, and Bagging Classifier. The models were evaluated using Accuracy, F1-score, confusion matrix, and ROC-based analysis. The study indicated that AI-enabled customer-support systems could improve customer engagement and service automation. However, the approach remained dependent on conventional predictive modelling and did not incorporate Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), or knowledge-grounded response generation.

Similarly, Singh (2022), in "Revolutionizing telecom customer support: The impact of AI on troubleshooting and service efficiency," examined the application of Artificial Intelligence to telecom customer support, particularly in troubleshooting and service efficiency. The study considered AI, predictive analytics, NLP, chatbots, and virtual assistants as mechanisms for improving customer-support operations. The reported findings indicated improvements in troubleshooting efficiency, response time, and customer satisfaction. However, the approach primarily represented conventional AI-based customer-support systems and did not investigate advanced LLM-based conversational systems or Retrieval-Augmented Generation.Table 2.1 below shown as the multiple papers and approches for NLP based for telecom customer support.

Table 2.1 – Summary of NLP-Based Approaches for Telecom Customer Support

Study / Year	Paper	Approach	Application	Key Methodology	Reported Findings	Research Gap
Patel and Trivedi (2020)	Leveraging predictive modeling, machine learning personalization, NLP customer support, and AI chatbots to increase customer loyalty	ML + NLP + Chatbot	Customer support and loyalty	Logistic Regression, KNN, SVC, Decision Tree, Random Forest, AdaBoost, Extra Trees and Bagging	Improved customer engagement and automated support	No LLM, RAG or knowledge-grounded generation
Singh (2022)	Revolutionizing telecom customer support: The impact of AI on troubleshooting and service efficiency	AI + NLP + Chatbot	Telecom troubleshooting	NLP, predictive analytics, chatbots and virtual assistants	Improved troubleshooting efficiency, response time and customer satisfaction	Limited investigation of LLMs and RAG
Yilma et al. (2025)	TelecomRAG: Taming Telecom Standards with Retrieval Augmented Generation and LLMs	RAG + LLM	Telecom question answering	Telecom knowledge retrieval and LLM generation	Improved domain-specific and knowledge-grounded responses	Primarily focused on telecom standards

2.3.4 Deep Learning Approaches

The adoption of deep learning has expanded AI applications in telecommunications by enabling automatic learning of complex feature representations from large customer datasets. Fujo et al. (2022) applied Deep-BP-ANN for telecom churn prediction, while Amatare and Ojo (2020) investigated CNN-based churn prediction. ANN models capture nonlinear relationships, whereas CNNs identify local patterns in customer data. Sequential architectures, including RNN, LSTM, GRU, and BiLSTM, further support contextual sequence modelling; however, the current thesis

reference list does not provide specific telecom customer-support experiments for these architectures. Alkhayrat et al. (2020) demonstrated deep learning for telecom customer segmentation using the SyriaTel dataset. Overall, conventional deep learning remains focused mainly on prediction, classification, and feature learning rather than external knowledge retrieval. These limitations motivate the transition towards Transformer architectures, LLMs, semantic retrieval, and RAG for automated telecom customer support. Table 2.2 summarises the deep learning approaches in telecommunications.

Table 2.2 – Summary of Deep Learning Approaches in Telecom

Study / Year	Model Type	DataSets	Key Methodology	Application	Key Limitation
Amatare & Ojo (2020)	CNN	Telecom customer churn dataset*	CNN-based automatic feature learning	Customer churn prediction	Focused on prediction rather than conversational support
Fujo et al. (2022)	Deep-BP-ANN	IBM Telco Customer Churn – 7,043; Cell2Cell – 51,047	Deep neural network compared with XGBoost, LR, NB and KNN	Customer churn prediction	No conversational AI, RAG or real-time knowledge retrieval
Alkhayrat et al. (2020)	Autoencoder + PCA	SyriaTel Telecom – 100,000 records, 220 features	Dimensionality reduction and clustering	Customer segmentation	No NLP, conversational support or RAG
RNN	Recurrent Neural Network	Not specified in current references	Sequential representation learning	Sequential text/conversation	No supporting telecom study in current references
LSTM	Long Short-Term Memory	Not specified in current references	Long-term dependency modelling	Conversational/sequential text	No supporting telecom study in

					current references
GRU	Gated Recurrent Unit	Not specified in current references	Lightweight gated sequence modelling	Sequential text	No supporting telecom study in current references

2.3.5 Transformer-Based Approaches

Transformer-based approaches represent a major transition towards modern language understanding and knowledge-grounded AI. Unlike RNNs and LSTMs, Transformers use self-attention to capture contextual relationships, supporting BERT, RoBERTa, DistilBERT, Sentence Transformers, and LLMs. BERT supports intent classification and semantic matching, while Sentence Transformers generate embeddings for dense retrieval. Erak et al. (2024) investigated long-context RAG for 3GPP standards, while Yilma et al. (2025) proposed TelecomRAG using 3GPP Release 16 and 18. Bornea et al. (2025) introduced Telco-oRAG using hybrid retrieval and neural routing, and Yuan et al. (2025) explored knowledge-graph integration. These approaches support telecom applications including intent classification, semantic retrieval, reranking, and knowledge grounding. The proposed framework applies these concepts using vector databases such as FAISS and ChromaDB.

2.3.6 Large Language Models for Customer Support

The emergence of Large Language Models (LLMs) has transformed automated customer support by enabling natural-language understanding, contextual conversation, summarisation, and response generation. Models such as GPT, LLaMA, Claude, and Gemini provide capabilities beyond conventional rule-based chatbots and traditional NLP systems. However, standalone LLMs may produce outdated, inaccurate, or unsupported information, creating challenges for domain-specific telecom applications. Recent research demonstrates the transition towards specialised telecom solutions. Yilma et al. (2025) proposed TelecomRAG using 3GPP Release 16 and 18 standards, while Bornea et al. (2025) introduced Telco-oRAG using hybrid retrieval and neural routing. Yuan et al. (2025) investigated knowledge graphs, dense retrieval, and RAG for improved grounding. Analytics and Sukhwal (2024) evaluated customer-support RAG using MRR

and Recall@K. These findings support the proposed framework integrating trusted telecom knowledge, semantic retrieval, vector databases, prompt engineering, and LLM-based generation.able 2.3 summarises commonly used LLM-based approaches for telecom customer support, highlighting their key characteristics, applications, and contributions to improving automated customer service.

Table 2.3 – Summary of LLM-Based Approaches for Telecom Customer Support

Study / Year	**LLM / Approach**	**Dataset / Knowledge Source**	**Key Methodology**	**Application**	**Main Limitation**
Bornea et al. (2025)	LLM + Telco-oRAG	Telecom query/knowledge data	Hybrid retrieval + neural routing	Telecom queries	Further scalability and enterprise deployment research required
Yuan et al. (2025)	LLM + KG-RAG	Telecommunications-domain documents	Knowledge graphs + dense retrieval + RAG	Telecom QA	Limited real-world customer-support evaluation
Analytics & Sukhwal (2024)	LLM + RAG chatbot	Customer-support dataset	Keyword extraction + document retrieval + prompt engineering	Customer support	Mainly retrieval-focused evaluation
LLaMA	Open-weight LLM family	General pre-trained knowledge	Generative language modelling	Enterprise/custom applications	Requires domain grounding for telecom
Gemini	Multimodal/general LLM family	General pre-trained knowledge	Language understanding and generation	Conversational AI	Requires domain-specific grounding

2.4 Overview of Telecom-Specific RAG: Telecom-specific Retrieval-Augmented Generation (RAG) combines Large Language Models (LLMs) with external domain knowledge to generate accurate, relevant, and context-aware responses. Unlike standalone LLMs, RAG retrieves relevant information before generating an answer, helping reduce hallucinations and improve factual reliability. In telecommunications, this approach is particularly useful for handling complex customer queries involving billing, network troubleshooting, service plans, technical support, and domain-specific terminology. Telecom-specific RAG can integrate semantic retrieval, vector databases, knowledge graphs, and document-based knowledge sources to provide grounded responses. This approach supports more reliable, scalable, and context-aware automated customer-support systems.

2.4.1 Evolution of Telecom RAG Systems

Evaluation of Telecom RAG systems is essential for determining whether retrieved information is relevant, responses are accurate, and generated answers remain grounded in authoritative telecom knowledge. Roychowdhury et al. (2024) examined evaluation metrics for telecom question answering, highlighting the importance of assessing retrieval and response quality. The proposed research evaluates Telecom RAG using Precision@5, Recall@K, Mean Reciprocal Rank (MRR), Answer Accuracy, Response Relevance, Grounding, Latency, and Query Resolution. These metrics assess different stages of the RAG pipeline, from document retrieval to final response generation. Analytics and Sukhwal (2024) used MRR and Recall@K for customer-support RAG evaluation. The framework therefore provides a broader evaluation covering retrieval effectiveness, response quality, factual grounding, efficiency, and customer-support resolution.

2.4.2 Vector Databases for Telecom RAG

Vector databases are essential to RAG because they enable efficient storage, indexing, and semantic retrieval of domain-specific information. Telecom FAQs, troubleshooting manuals, billing documents, service information, and 3GPP standards can be embedded and stored in FAISS or ChromaDB. Yilma et al. (2025) demonstrated telecom retrieval using 3GPP standards, while Bornea et al. (2025) explored hybrid retrieval and neural routing. Yuan et al. (2025) investigated

knowledge-graph integration, and Roychowdhury et al. (2024) examined telecom RAG evaluation. Therefore, the vector database acts as the semantic knowledge layer between telecom information and the LLM, enabling grounded response generation and evaluation through retrieval and response-quality metrics.

2.4.3 Hybrid, Semantic, Reranking and 3GPP Knowledge Retrieval:

Hybrid, semantic, reranking, and 3GPP retrieval are important components of telecom-specific RAG systems. Hybrid retrieval combines sparse keyword search with dense semantic retrieval, capturing both exact telecom terms and contextual meaning. Bornea et al. (2025) proposed Telco-oRAG using hybrid retrieval and neural routing to improve retrieval precision and response quality. Semantic retrieval supports FAISS and ChromaDB through embeddings, while reranking prioritises relevant documents and reduces irrelevant LLM context. Analytics and Sukhwal (2024) evaluated retrieval using MRR and Recall@K. Erak et al. (2024) and Yilma et al. (2025) demonstrate 3GPP-based RAG approaches. Overall, the proposed pipeline integrates hybrid retrieval, semantic search, reranking, 3GPP knowledge retrieval, and LLM generation for grounded telecom customer support. Figure-2.3 illustrates a conceptual Hybrid Semantic Retrieval and Retrieval-Augmented Generation (RAG) Pipeline for Telecom Customer Support

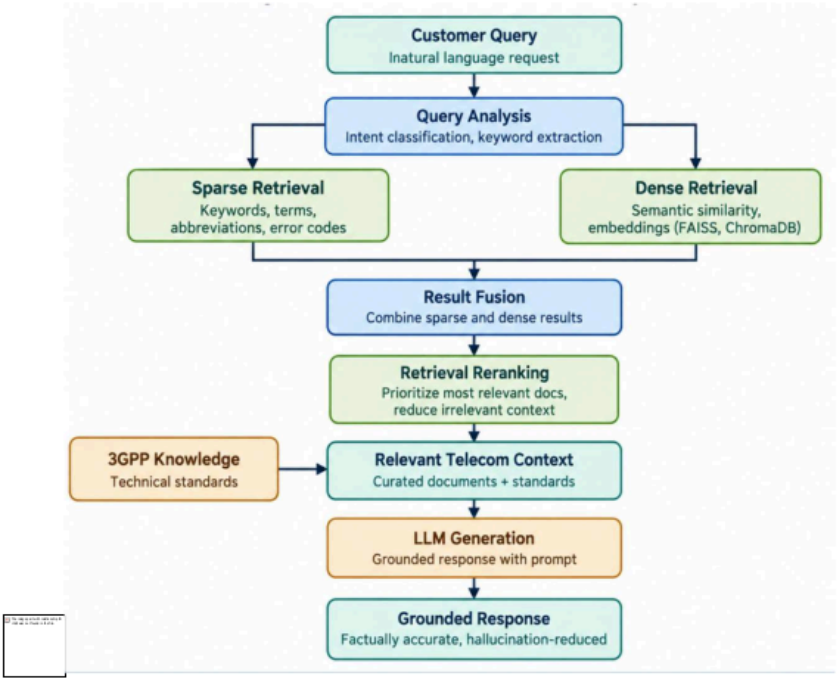


Figure 2.3- Hybrid sementic Retrieval & RAG Pipeline

2.5 Transformer-Based Approaches for Telecom Customer Support

Transformer-based architectures overcome limitations of RNNs and LSTMs through self-attention, enabling effective understanding of long-range contextual relationships and telecom-specific terminology. They support intent classification, semantic similarity, question answering, embeddings, and information retrieval. Their contextual representations enable semantic and hybrid retrieval within RAG systems, providing the technological foundation for Generative AI for RAG-Based AI System for Automated Telecom Customer Support and supporting context-aware, knowledge-grounded response generation.

2.5.1 Evolution of Transformer Architecture

Transformer architecture advanced NLP by overcoming the sequential limitations of RNNs and LSTMs through self-attention and multi-head attention. These mechanisms capture contextual relationships while enabling parallel processing, forming the foundation of BERT, RoBERTa, DistilBERT, Sentence Transformers, and modern LLMs. In telecom support, Transformers enable semantic understanding of technical terminology and context-dependent queries. Erak et al. (2024) explored long-context RAG for 3GPP standards, while Yilma et al. (2025) developed TelecomRAG using 3GPP Releases 16 and 18. Roychowdhury et al. (2024) examined telecom RAG evaluation, and Sousa et al. (2024) investigated document organisation within RAG. Thus, Transformers provide the architectural foundation for semantic retrieval and knowledge-grounded telecom customer support.

2.5.2 BERT based Approaches

BERT improved contextual language understanding by considering both directions of a sentence. This is relevant to telecom support because context can distinguish intents such as roaming and connectivity. BERT can support intent classification, semantic matching, and question answering before retrieval and response generation. Yilma et al. (2025) and Erak et al. (2024) demonstrate telecom RAG applications using 3GPP standards. However, the reviewed literature does not directly evaluate BERT on a telecom customer-support dataset. Transformer architectures such as RoBERTa, DistilBERT, and ALBERT were developed to improve BERT's training efficiency,

representation quality, and computational requirements. These models are relevant to telecom customer support, where low-latency and scalable query processing are important. RoBERTa can support contextual query understanding, while DistilBERT and ALBERT offer efficiency advantages for intent classification and routing. Telecom studies such as Yilma et al. (2025), Erak et al. (2024), and Bornea et al. (2025) demonstrate the importance of effective retrieval and knowledge grounding, although they do not directly benchmark these Transformer variants. Therefore, the current thesis positions RoBERTa, DistilBERT, and ALBERT as potential lightweight components for query understanding within a broader telecom RAG architecture rather than as independently validated telecom models.

2.5.3 Transformer-Based Intent Classification

Intent classification is an important component of automated telecom customer support because queries must be understood before relevant knowledge is retrieved. Traditional NLP approaches relied mainly on keywords and predefined categories, whereas Transformer-based models use contextual representations to identify underlying meanings. For example, different expressions such as “My internet has stopped working” and “Mobile data isn't connecting” may represent the same connectivity intent. Patel and Trivedi (2020) demonstrated the usefulness of conventional machine-learning and NLP classification for customer support, while Singh (2022) explored AI, NLP, chatbots, and virtual assistants for telecom troubleshooting. More recent research, including Bornea et al. (2025), connects query understanding with hybrid retrieval and neural routing. Analytics and Sukhwal (2024) further demonstrated keyword extraction and retrieval for customer-support RAG. These studies support intent-aware routing within the proposed telecom RAG framework.

2.5.4 Transformer-Based Question Answering

Transformer-based question answering is important for telecom customer support because many queries require domain-specific information beyond general language knowledge. Telecom questions may involve technical standards, network procedures, configuration, troubleshooting, billing, and roaming. Transformer models provide contextual understanding, while RAG retrieves relevant information before generating responses. Erak et al. (2024) investigated fine-tuned RAG with long-context support for 3GPP standards, while Yilma et al. (2025) developed TelecomRAG

using 3GPP Release 16 and 18. Roychowdhury et al. (2024) highlighted the importance of evaluating both retrieval and answer quality. Similarly, Sousa et al. (2024) explored document organisation using Bisecting K-Means, and Yuan et al. (2025) investigated knowledge-graph-enhanced RAG. These studies collectively support the proposed end-to-end framework combining semantic retrieval, vector databases, reranking, and LLM-based generation.

2.5.5 Transformer-Based Retrieval and Reranking

Transformer-based retrieval is central to telecom RAG because systems must identify relevant information from large document collections. Sentence Transformers convert customer queries and documents into semantic embeddings, enabling contextual retrieval through vector databases such as FAISS and ChromaDB. For example, a query about losing mobile connectivity while travelling abroad can retrieve documents discussing international roaming, even when exact keywords differ. Yilma et al. (2025) demonstrated telecom RAG using 3GPP Release 16 and 18 standards, while Erak et al. (2024) examined long-context RAG for lengthy 3GPP documents. Bornea et al. (2025) introduced Telco-oRAG using hybrid retrieval and neural routing. Reranking can further prioritise relevant evidence before LLM generation. Yuan et al. (2025) additionally explored knowledge-graph-enhanced RAG, demonstrating increasingly sophisticated telecom retrieval approaches.

2.5.6 Multilingual Transformers

Multilingual Transformers extend Transformer-based language understanding across languages, making them potentially valuable for telecom customer support in diverse markets. They can help identify semantically similar customer intents expressed through different languages, regional expressions, transliteration, or mixed-language queries. For example, queries such as “My internet is not working” expressed in different languages could potentially be mapped to similar semantic representations and connected to the same telecom knowledge. However, multilingual Transformers remain outside the core scope of this study, while multilingual speech processing is explicitly excluded. The current thesis does not provide separate empirical telecom results for multilingual Transformers. Future research could extend the proposed knowledge repository using TelecomRAG, 3GPP standards, FAQs, troubleshooting manuals, and customer-support data.

Studies by Yilma et al. (2025), Bornea et al. (2025), and Yuan et al. (2025) provide relevant foundations for future multilingual telecom RAG research.

2.6 Research Gaps

Despite advances in Artificial Intelligence, NLP, LLMs, and RAG for telecommunications, several research gaps remain. Existing studies demonstrate domain-specific retrieval, technical question answering, hybrid retrieval, and knowledge grounding, but these components are often investigated separately rather than within a comprehensive end-to-end customer-support framework. These gaps motivate the proposed research,"Generative AI for RAG-Based AI System for Automated Telecom Customer Support. First, existing research focuses mainly on telecom standards and technical question answering. Yilma et al. (2025) developed TelecomRAG using 3GPP Release 16 and 18, while Erak et al. (2024) examined long-context RAG for 3GPP standards. Although valuable, these approaches do not fully address broader customer-support interactions. Second, limited research evaluates complete pipelines covering query understanding, retrieval, reranking, generation, and customer-oriented outcomes. Sukhwal (2024) primarily evaluated retrieval using MRR and Recall@K, creating scope for broader evaluation of answer accuracy, response relevance, hallucination, and resolution time.

Third, retrieval optimisation remains an open challenge.Bornea et al. (2025) demonstrated hybrid retrieval and neural routing through Telco-oRAG, but comparative evaluation across dense retrieval, hybrid retrieval, knowledge graphs, and reranking remains limited. Fourth, heterogeneous telecom knowledge sources including standards, FAQs, troubleshooting guides, billing information, and service documentation require structured integration through vector databases and semantic retrieval. Fifth, stronger evaluation of grounding and hallucination reduction is required. Yuan et al. (2025) demonstrated knowledge graphs with dense retrieval and RAG, but broader customer-support evaluation remains limited. Finally, privacy, security, transparency, fairness, explainability, and governance require greater consideration for enterprise deployment. Multilingual support is retained as a future research direction rather than a component of the current study. These gaps justify an integrated framework combining LLMs, telecom knowledge retrieval, vector databases, hybrid retrieval, reranking, knowledge grounding, and responsible-AI evaluation.Table 2.4 illustrate that key research gaps identified in the existing literature and the corresponding contributions of the present study are summarised

Table 2.4- Summary of Research Gaps and Contributions

Research Area	Limitations in Previous Work	Research Gap	Addressed in This Study
Traditional Customer Support	Rule-based and keyword systems struggle with complex and context-dependent queries.	Limited contextual understanding.	Develops LLM + RAG-based conversational support.
Telecom Knowledge Retrieval	TelecomRAG primarily focuses on 3GPP standards.	Limited coverage of broader customer-support knowledge.	Integrates TelecomRAG, 3GPP, FAQs, troubleshooting guides, billing and service documents.
LLM-Based Support	Standalone LLMs may generate outdated or hallucinated information.	Lack of reliable external knowledge grounding.	Grounds LLM responses using retrieved telecom evidence.
Vector Databases	Limited integrated investigation of semantic knowledge repositories for customer support.	Need for efficient telecom document indexing and retrieval.	Uses FAISS/ChromaDB for embedding storage and semantic retrieval.

2.7 Discussion

The literature demonstrates a clear transition from conventional machine-learning and rule-based telecom support towards Transformer-based, Large Language Model (LLM), and Retrieval-Augmented Generation (RAG) approaches. Earlier studies, including Patel and Trivedi (2020) and Singh (2022), established the value of NLP, predictive modelling, chatbots, and virtual assistants for telecom support. Recent research highlights domain-specific RAG, with TelecomRAG using 3GPP standards for telecom question answering (Yilma et al., 2025), while Telco-oRAG combines hybrid retrieval and neural routing to improve retrieval performance (Bornea et al., 2025). Yuan et al. (2025) further demonstrated the value of knowledge graphs and dense retrieval for structured

telecom knowledge grounding. Analytics and Sukhwal (2024) highlighted the use of MRR and Recall@K for customer-support RAG evaluation. These findings motivate the proposed framework integrating telecom knowledge sources, vector databases, hybrid retrieval, reranking, and LLM generation, evaluated through retrieval, accuracy, relevance, grounding, and resolution measures.

2.8 Summary

This chapter reviewed 30 relevant studies covering traditional machine learning, NLP, deep learning, Transformers, LLMs, telecom-specific RAG, vector databases, semantic and hybrid retrieval, reranking, and knowledge-grounded customer support. The literature demonstrates an evolution from conventional predictive models and rule-based chatbots towards context-aware Generative AI systems. Traditional ML and deep-learning approaches established foundations for telecom analytics but remain limited in understanding complex customer queries. Transformer architectures, Sentence Transformers, and LLMs enable contextual understanding and semantic retrieval through vector databases such as FAISS and ChromaDB. Telecom-specific studies, including Yilma et al. (2025), Bornea et al. (2025), Erak et al. (2024), and Yuan et al. (2025), demonstrate the value of RAG, hybrid retrieval, long-context retrieval, and knowledge graphs. Key gaps include limited end-to-end telecom RAG evaluation, insufficient grounding and hallucination assessment, and limited privacy and governance analysis. These gaps justify the proposed integrated telecom RAG framework.

CHAPTER 3 – RESEARCH METHODOLOGY

The methodological design of this study establishes the systematic procedures used to design, develop, implement, and evaluate a Generative AI and Retrieval-Augmented Generation (RAG)-based system for automated telecom customer support. Given the complexity and domain-specific nature of telecommunications queries, a structured research methodology is essential to ensure reproducibility, reliable evaluation, knowledge grounding, and effective comparison with existing customer-support approaches.

3.1 Introduction

This chapter presents the methodological framework adopted to design, develop, implement, and evaluate the proposed Generative AI for RAG-Based AI System for Automated Telecom Customer Support. The methodology integrates Natural Language Processing, semantic embeddings, vector database retrieval, hybrid information retrieval, reranking, prompt-based generation, and Large Language Models within a structured experimental framework. The use of Retrieval-Augmented Generation is particularly appropriate for telecommunications because customer-support queries can require domain-specific information that may not be reliably available from the pretrained knowledge of a general-purpose language model. Yilma et al. (2025) demonstrated the application of RAG to telecom standards through TelecomRAG, highlighting the importance of retrieving domain-specific information before language-model generation. The methodology progresses systematically from data selection and preprocessing to vector database deployment, RAG model development, model configuration, evaluation, performance comparison, optimisation, and customer-query response generation. Bornea et al. (2025) proposed Telco-oRAG, which combines hybrid retrieval and neural routing for telecom queries, supporting the methodological decision to investigate both semantic and hybrid retrieval strategies in the proposed system. The overall methodology therefore provides a controlled experimental framework for investigating retrieval effectiveness, response quality, grounding and computational efficiency. The below mentioned Figure 3.1 describes the framework of RAG based AI system for Automated Telecom Customer Support.



Figure 3.1 Framework of RAG based AI system for Automated Telecom Customer Support

3.2 Research Design:

This study adopts an experimental quantitative research design to develop and evaluate the proposed RAG-based telecom customer-support system. The experimental approach enables different retrieval and generation configurations to be implemented under controlled conditions and evaluated using common test queries and performance measures. The methodology investigates whether progressively advanced retrieval mechanisms can improve the quality, relevance and grounding of automated customer-support responses.

The experimental framework consists of four configurations: Traditional Retrieval, Vector Retrieval, Standard RAG and Optimised RAG. Traditional Retrieval provides a lexical baseline using TF-IDF, while Vector Retrieval uses semantic embeddings and FAISS. Standard RAG combines semantic retrieval with response generation, whereas Optimised RAG introduces hybrid retrieval, metadata filtering and reranking. This layered design is consistent with the research

direction identified in the thesis literature, where hybrid retrieval and reranking are considered important mechanisms for improving telecom retrieval quality. The evaluation considers both retrieval and generation performance. Analytics and Sukhwal (2024) evaluated a RAG-based customer-support chatbot using document retrieval, keyword extraction and prompt engineering, with MRR and Recall@K used to assess retrieval performance. This provides methodological support for evaluating both retrieval effectiveness and generated-response quality rather than assessing language fluency alone.

3.3 Data Selection:

Data selection is an important methodological stage because the quality and relevance of the underlying customer-support information directly influence retrieval and response-generation performance. The primary dataset selected for the experimental implementation is the Customer Support dataset, available through Kaggle. The dataset provides conversational customer-support interactions that can be transformed into customer queries and corresponding support responses. This makes the dataset appropriate for evaluating an automated system intended to understand customer requirements and generate knowledge-grounded responses. The same rationale is reflected in the existing methodology in your thesis.

3.4 Data Pre-Processing

Data preprocessing prepares the selected customer-support information for semantic retrieval, vector indexing and response generation. The supplied Python pipeline normalises common message-level and ticket-style schemas into a canonical structure, cleans HTML-like tags and repeated whitespace, handles missing values, removes empty messages, and preserves available metadata such as industry, product/category, issue type, locale/language, channel, sentiment, priority/urgency, status/outcome and intent. For message-level data, turns are ordered within conversations and customer-to-agent question–answer pairs are extracted from consecutive recognised roles. For ticket-style data, the query is constructed from available subject/body fields and paired with the supplied answer. The exact schema path used should be confirmed from the final run summary.

3.5 Model Selection

The experimental implementation compares four configurations: Traditional Retrieval (TF-IDF), Vector Retrieval, RAG, and Optimised Hybrid RAG. The TF-IDF baseline provides lexical retrieval. The dense retrieval path is configured for BAAI/bge-m3 embeddings and FAISS when those optional dependencies and model weights are available. The RAG path combines retrieval with response generation; Azure OpenAI is used only when the required environment variables are configured, otherwise the script may use an extractive fallback. The optimised path combines candidate retrieval, metadata filtering, and optional cross-encoder reranking. The actual backends loaded for the reported experiment should be confirmed from the run artefacts.

3.5.1 Vector Database Deployment

When the dense retrieval dependencies are available, the implementation creates BAAI/bge-m3 embeddings and indexes them with FAISS. Embeddings are normalised and an inner-product index is used for similarity search. If the embedding or FAISS stack is unavailable, the pipeline can use its configured fallback; therefore, the presence of BGE-M3/FAISS in the design does not alone prove that they were active in every run. Figure 3.2 define that Vector Database Deployment Architecture for BGE-M3 Embeddings and FAISS-Based Semantic Retrieval.

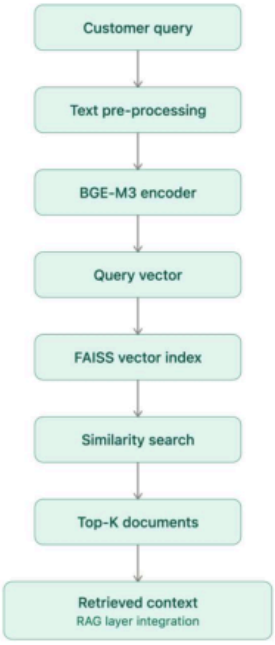


Figure 3.2– Vector Database Deployment Architecture

3.5.2 Rag Model Design and Development

The RAG workflow retrieves candidate reference answers and supplies retrieved context to the response-generation layer. Azure OpenAI generation is conditional on endpoint, API key and deployment configuration. When cloud generation is not configured or fails, an extractive fallback may be used. For this reason, model and generation backend information must be reported alongside all results. The proposed architecture contains multiple retrieval stages which is shown figure 3.3 about RAG model architecture. First, TF-IDF provides a traditional lexical retrieval baseline. Second, BGE-M3 and FAISS perform dense semantic retrieval. The optimised architecture combines these retrieval outputs into a candidate pool. Hybrid retrieval is motivated by the fact that sparse retrieval can identify exact telecom terminology, service names and technical terms, while dense retrieval can capture semantic similarity. Bornea et al. (2025) similarly investigated hybrid retrieval for telecom queries. After candidate retrieval, metadata filtering is applied where relevant product and issue information is available. A cross-encoder reranker then evaluates query-document relationships and reorders the retrieved candidates. Reranking is intended to reduce irrelevant context supplied to the generation model and improve the relevance of the evidence used for response generation. Your thesis literature explicitly identifies reranking as a mechanism for improving retrieval relevance and factual grounding. A Sentence Transformer-based embedding model can be used to convert customer queries and telecom documents into dense vectors. This component is particularly important because the resulting embeddings are stored in the vector database for semantic retrieval.

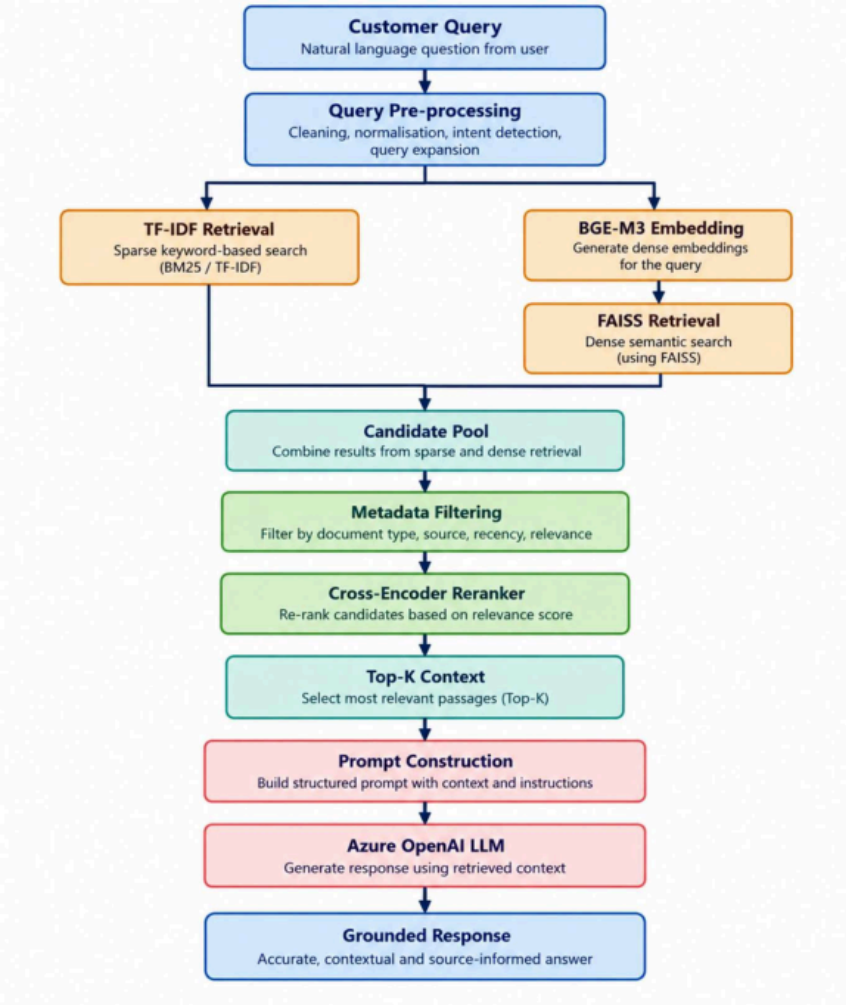


Figure 3.3- Proposed RAG Model Architecture

3.6 Model Configuration and Experimental Setup

The proposed system uses multiple model configurations rather than relying on a single model. This enables the study to determine the incremental contribution of traditional retrieval, semantic retrieval, RAG generation, hybrid retrieval and reranking. The first configuration, Traditional Retrieval, uses TF-IDF and cosine similarity. The second, Vector Retrieval, uses BGE-M3 embeddings and FAISS. The third, Standard RAG, combines BGE-M3/FAISS retrieval with RAG-based response generation. The fourth, Optimized RAG, combines TF-IDF and dense retrieval, metadata filtering, cross-encoder reranking and RAG generation. The implementation directly reflects these four configurations. The experimental code explicitly implements

Traditional Retrieval, Vector Retrieval, RAG and Optimized RAG configurations. The embedding model is configured as BAAI/bge-m3, while the reranking component uses BAAI/bge-reranker-v2-m3. The standard retrieval configuration uses Top-K = 5, while the optimized architecture retrieves a larger candidate set before reranking. It is important to distinguish model configuration from model training. The current implementation uses pretrained embedding and reranking models and does not perform fine-tuning of BGE-M3 or the LLM. Therefore, the methodological terminology "Model Configuration and Experimental Setup" accurately reflects the implemented research procedure.

3.7 Model Evaluation and Performance Analysis

The evaluation methodology measures both retrieval effectiveness and response-generation performance. This is important because a RAG system may generate fluent responses while retrieving inappropriate evidence. Roychowdhury et al. (2024) specifically investigated evaluation metrics for question answering in the telecom domain and emphasised the importance of evaluating RAG systems beyond language fluency. Retrieval performance is measured using Precision@5, Recall@5 and MRR. The implementation assigns relevance using product and issue-type matching between retrieved records and the evaluation record. Therefore, these measures should be described as metadata-based relevance proxies, rather than human-annotated gold relevance judgements. Generated responses are evaluated using Answer Accuracy, Response Relevance and Grounding. Answer Accuracy uses lexical F1 against the reference answer. Response Relevance uses embedding similarity between the query and generated answer, while Grounding measures overlap between generated content and retrieved context. The implementation records these measures for Standard RAG and Optimised RAG. Latency is measured in milliseconds using execution-time measurements. This is particularly important for automated customer-support systems because retrieval and generation must be sufficiently efficient for practical use.

3.8 Comparison and Performance Optimization

The pipeline compares retrieval and response-quality proxies, grounding proxy, and latency across the four configurations. It normalises higher-is-better metrics and inverse-normalises latency to calculate a composite score. The implementation explicitly treats latency as a performance measure and not as actual cloud/API cost; cost must be measured separately using token usage and infrastructure billing. Implementation methodology- idefines the aggregation and comparison procedure used to calculate model-level performance. Performance optimisation is achieved by combining sparse and dense retrieval. The candidate documents are subsequently filtered using metadata where available and reranked using a cross-encoder. This methodology is supported by Bornea et al. (2025), whose Telco-oRAG approach combines hybrid retrieval and neural routing to improve retrieval precision and response quality. The implementation then calculates a weighted composite score using retrieval, answer quality, relevance, grounding and latency. Precision@5, Recall@5 and MRR each contribute 10%, Answer Accuracy 20%, Response Relevance 15%, Grounding 20% and Latency 15%.

3.9 Justification of Model Selection

Model selection was based on the functional requirements of each stage of the proposed RAG pipeline. Rather than using one model for all operations, the methodology employs complementary components for lexical retrieval, semantic representation, vector indexing, reranking and response generation.

TF-IDF was selected as the traditional retrieval baseline because it provides a transparent lexical retrieval mechanism against which semantic retrieval can be compared. This baseline is useful for determining whether the additional computational complexity of dense retrieval provides measurable improvements.BGE-M3 was selected as the semantic embedding component because Sentence Transformer-based representations allow customer queries and telecom knowledge to be represented in a shared vector space. The thesis identifies Sentence Transformers as an important foundation for semantic retrieval and vector databases. Yilma et al. (2025) and related RAG literature in the thesis- supports semantic embedding and retrieval as foundations of telecom. The cross-encoder reranker was selected because initial retrieval may return multiple semantically related candidates, and a second-stage relevance model can reassess the relationship between the query and candidate documents. This approach is consistent with the thesis discussion

of retrieval reranking for improving context relevance and grounding. GPT-based generation was selected for the final response layer because LLMs can transform retrieved telecom context into conversational natural-language responses. Your thesis identifies GPT, LLaMA, Claude and Gemini as candidate general-purpose LLM families and recommends selecting a primary model according to availability, computational requirements, deployment constraints, quality and cost. Model-selection discussion – identifies general-purpose LLMs as response-generation candidates and recommends selection based on quality, availability, deployment and cost.Table 3.1 provides a comparative summary of the selected model architectures, outlining their respective roles, underlying techniques, advantages, and relevance to the proposed telecom RAG framework.

Table 3.1 – Comparative Summary of Selected Model Architectures

Model / Approach	Function	Key Capability	Role in Study
TF-IDF	Lexical retrieval	Keyword-based similarity	Traditional baseline
BGE-M3	Semantic embedding	Dense semantic representation	Vector retrieval
FAISS	Vector indexing	Similarity search	Retrieval layer
BGE-Reranker-v2-M3	Reranking	Query-document relevance	Retrieval optimisation
Azure OpenAI / GPT	Response generation	Natural-language generation	RAG response layer

3.10 Evalution Metrics: The evaluation framework incorporates retrieval, response-quality and efficiency metrics because the effectiveness of a RAG system depends on both the quality of retrieved evidence and the quality of the final generated response. Roychowdhury et al. (2024) specifically investigated RAG evaluation in telecom question answering, providing support for evaluating multiple dimensions of RAG performance. Roychowdhury et al. (2024) – provides a telecom-specific evaluation perspective for RAG question answering.

1. Precision@5 : Implementation retrieves the top 5 documents for the traditional and vector retrieval models. The relevance label is 1 when the retrieved document has the same product and issue_type as the test query; otherwise it is 0.

$$Precision@5 = \frac{\sum_{i=1}^{5} Rel_i}{5} \quad (1)$$

where:

$$Rel_i = \begin{cases} 1, if Product_i = Product_q \wedge Issue_i = Issue_q \\ 0, otherwise \end{cases}$$

Therefore, if 4 of the top 5 retrieved documents are considered relevant: Precision@5=4/5=0.80

2. **Recall@5**: Implementation calculates the number of relevant retrieved documents and divides it by the total number of relevant documents available in the training set for the same product and issue type.

$$\textbf{Recall@5} = \frac{\sum_{i=1}^{5} Rel_i}{N_{relevant}} \quad (2)$$

where:

- $\sum R\,el_i$ = relevant documents retrieved in the top 5
- $N_{relevant}$ = total relevant training documents having the same product and issue type as the test query.

$$N_{relevant} = \sum_{j=1}^{N} \mathbb{I}\left(Product_j = Product_q \wedge Issue_j = Issue_q\right) \quad ()$$

where $\mathbb{I}(\cdot)$ is an indicator function.

3. **MRR(Mean Reciprocal Rank):** Implementation identifies the position of the first relevant retrieved document. If the first relevant document appears at rank rr, the reciprocal rank is 1/r1/r. If no relevant document is found, the score is 0.

For an individual query:

$$RR = \begin{cases} \frac{1}{r}, & \text{if the first relevant result is at rank } r \\ 0, & \text{if no relevant result exists} \end{cases}$$

Across Q queries:

$$MRR = \frac{1}{Q}\sum_{q=1}^{Q} R\,R_q \quad (3)$$

4. Response Relevance

Our implementation uses the BGE-M3 embedding model to encode both the customer query and generated answer. Because the embeddings are normalized, their dot product corresponds to cosine similarity. The Response Relevance is given below:

$$\text{Response Relevance} = E_q \cdot E_a \quad (4)$$

where:

- E_q = normalized BGE-M3 embedding of the query
- E_a = normalized BGE-M3 embedding of the generated answer.

Because:

$$\|E_q\| = \|E_a\| = 1 \quad (5)$$

the dot product is equivalent to cosine similarity:

$$\textbf{Response Relevance} = \frac{E_q \cdot E_a}{\|E_q\| \|E_a\|}. \quad (6)$$

and therefore:

$$\textbf{Response Relevance} = E_q \cdot E_a$$

5. Grounding Score: Implementation measures how many unique words in the generated answer also occur in the retrieved context.

- AA = unique words in generated answer , CC = unique words in retrieved context.
- $\textbf{Grounding} = \frac{|A \cap C|}{|A|}$

Therefore, if the generated answer contains 100 unique words and 85 also occur in the retrieved context:

- $\textbf{Grounding} = \frac{85}{100} = 0.85 \quad (7)$

8. TF-IDF Retrieval Similarity: Traditional retrieval baseline uses TF-IDF with:

- unigram and bigram features
- max_features = 150000
- sublinear_tf=True

and ranks documents using cosine similarity, The general cosine similarity used by the implementation is:

$$Cosine(Q, D_i) = \frac{Q \cdot D_i}{\|Q\| \|D_i\|} \quad (8)$$

where QQ is the TF-IDF query vector and DiD_i is the TF-IDF document vector. Documents are ranked in descending order of this score.

9.BGE-M3 + FAISS Vector Similarity: Our vector retrieval uses:
Embedding Model=BAAI/bge−m3

The embeddings are normalized before being stored.

FAISS uses: IndexFlatIP

which performs inner-product similarity, Because your embeddings are normalized:

$$Similarity(Q, D_i) = E_Q \cdot E_{D_i} \quad (9)$$

and this is equivalent to:

$$Similarity(Q, D_i) = Cosine(E_Q, E_{D_i}) \quad (9)$$

The top K=5K=5 documents are selected according to the highest similarity scores.

10. Candidate Retrieval in Optimized RAG:

$$K_c = 10$$

Candidates from each retrieval mechanism:

$$C_{TFIDF} = TopK_{10}(TFIDF(Q))$$

$$C_{BGE} = TopK_{10}(BGE(Q))$$

The candidate pool is then formed as the union of both sets:

$$C = C_{TFIDF} \cup C_{BGE} \quad (10)$$

Duplicate documents are removed by the implementation.

11. Metadata Filtering: Implementation applies product and issue-type filtering when sufficient matching documents exist.

The allowed document set is:

$$D_{allowed} = \{d_i : Product_i = Product_q \wedge Issue_i = Issue_q\} \quad (11)$$

If at least 20 matching documents exist and enough filtered candidates remain, the candidate set is replaced by the filtered candidates.

12. Cross Encoder Reranking: Our Implementation uses:

$$BAAI/bge - reranker - v2 - m3$$

The reranker receives query-document pairs:

$$(Q, D_i)$$

and assigns each pair a relevance score:

$$S_i = Reranker(Q, D_i)$$

The candidates are sorted in descending order of S_i, and the top:

$$K_{opt} = 3 \tag{12}$$

are selected.

13. Composite Model Score:

Code first performs min-max normalization on the metrics where higher is better.

For a higher-is-better metric:

$$M_{norm} = \frac{M - M_{min}}{M_{max} - M_{min}}$$

Therefore:

$$Precision_{norm} = \frac{Precision - P_{min}}{P_{max} - P_{min}}. \tag{13}$$

and similarly for Recall, MRR, Answer Accuracy, Response Relevance and Grounding.

14. Latency Normalization: Latency is different because lower is better, implementation therefore uses:

$$\boxed{\boldsymbol{Latency_{norm} = \frac{Latency_{max} - Latency}{Latency_{max} - Latency_{min}}}} \tag{14}$$

Thus:

- lowest latency → 1
- highest latency → 0.

3.11 Experimental Environment

The experimental environment consists of Python-based machine-learning and natural-language-processing tools required to implement the complete retrieval and generation pipeline. The implementation uses Pandas, NumPy, Scikit-learn, Sentence Transformers, FAISS and Matplotlib, together with an optional Azure OpenAI integration. The use of vector databases and semantic retrieval is consistent with the technical architecture described in the thesis, which identifies FAISS and ChromaDB as suitable mechanisms for storing and retrieving telecom knowledge embeddings. The experimental configuration uses a fixed random seed of 42, Top-K retrieval of 5, candidate retrieval of 10, and optimized Top-K retrieval of 3. The BGE-M3 embedding model and BGE reranker are loaded as pretrained models. The methodology also records model comparison results and predictions for subsequent analysis. This supports reproducibility by maintaining consistent evaluation queries, configuration parameters and output artefacts.

3.12 Customer query resolution and response Generation

Customer query resolution and response generation represents the final stage of the offline experiment. An incoming query is normalised, relevant reference answers are retrieved, and the selected context is passed to the generation layer when configured. The response is then compared with the reference answer and evaluated using the metrics available in the pipeline. The prototype does not execute customer-account changes or telecom network operations.Figure 3.4 illustrates the customer query resolution flow, outlining the key stages of query normalisation, reference answer retrieval, response generation, and evaluation within the proposed telecom RAG framework.

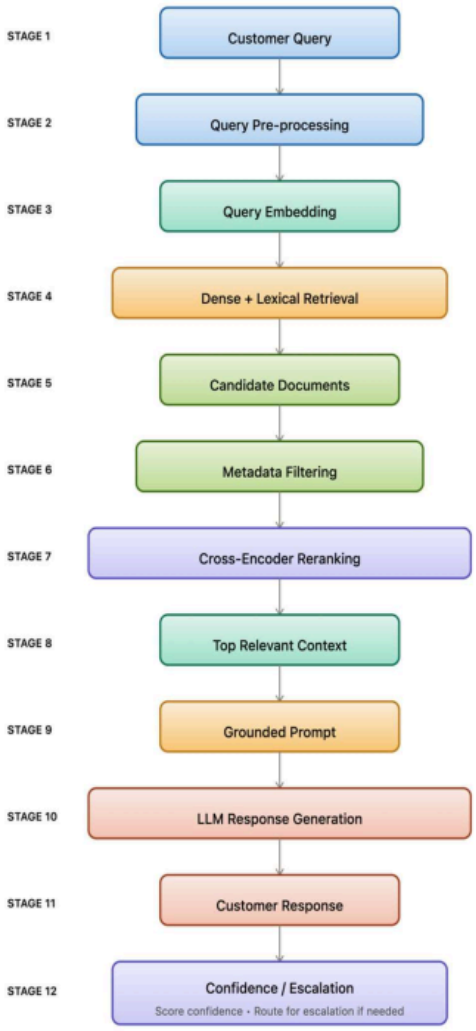


Figure 3.4 – Customer Query Resolution Flow

3.13 Summary with System Configuration Details

The complete research methodology for developing and evaluating the proposed Generative AI and RAG-based automated telecom customer-support system. The methodology begins with customer-support data selection and preprocessing, followed by conversational QA construction and conversation-level dataset separation. The vector retrieval layer uses BGE-M3 embeddings and FAISS, while the optimised retrieval architecture combines TF-IDF, dense retrieval, metadata filtering and cross-encoder reranking. The methodology is supported by telecom RAG research including TelecomRAG by Yilma et al. (2025), hybrid retrieval research by Bornea et al. (2025), long-context telecom RAG by Erak et al. (2024), and RAG evaluation research by Roychowdhury

et al. (2024). These studies collectively support the methodological focus on domain-specific retrieval, hybrid search, contextual grounding and multi-dimensional evaluation. Figure 3.2 presents the system configuration details, summarising the experimental approach, dataset, embedding model, vector index, reranking mechanism, generation layer, retrieval parameters, and evaluation metrics used to develop and assess the proposed telecom RAG framework.

Figure: 3.2- System Configuration Details

Component	Configuration
Research approach	Experimental / Quantitative
Dataset	Customer Support Conversations
Domain	Telecommunications
Retrieval baseline	TF-IDF
Embedding model	BAAI/bge-m3
Vector index	FAISS
Reranker	BAAI/bge-reranker-v2-m3
Generation	Azure OpenAI
Standard Top-K	5
Candidate-K	10
Optimised Top-K	3
Random seed	42
Retrieval metrics	Precision@5, Recall@5, MRR
Response metrics	Answer Accuracy, Relevance, Grounding
Efficiency metric	Latency
Evaluation models	Traditional Retrieval, Vector Retrieval, RAG, Optimised RAG

CHAPTER 4- ANAYLSIS

4.1 Introduction

This chapter presents a detailed analysis of the dataset used to develop and evaluate the proposed Generative AI for RAG-Based AI System for Automated Telecom Customer Support. The purpose of the analysis is to understand the structure, composition, linguistic characteristics, and metadata distribution of the customer-support data before applying the retrieval and generation framework

described in Chapter 3. The analysis focuses on the transformation of a large mixed-industry customer-support dataset into a controlled telecom-specific research dataset. Particular attention is given to the distribution of telecom products, support issues, language, sentiment, urgency, message length, and customer-query/reference-answer characteristics. Such analysis is important because the quality and diversity of the underlying data can directly influence retrieval effectiveness and the quality of generated responses in RAG systems (Lewis et al., 2020; Yilma et al., 2025). The chapter is organised into four major components: Dataset description and composition., Data preparation and creation of customer–agent question-answer pairs,Exploratory Data Analysis (EDA) of telecom customer-support interactions. Interpretation of the observed data characteristics and their implications for the proposed RAG architecture. The findings presented in this chapter provide the empirical foundation for the experimental results and model comparison presented in Chapter 5.

4.2 Dataset Description

The primary experimental dataset used in this research is a large customer-support conversation dataset containing interactions across multiple industries. The implementation first performs a dataset audit and subsequently isolates records belonging to the telecommunications domain. Because the original dataset contains mixed-industry records and some combinations of products and issue labels that are not strictly telecom-specific, a further filtering operation was introduced to create a strict telecom subset. The source dataset contains 976,271 interaction records distributed across 69,731 conversations and 16 attributes. The available attributes include conversation identifier, turn index, role, message text, timestamp, industry, product, issue type, language, communication channel, customer and agent identifiers, sentiment, urgency, outcome, and primary intent. Table 4.1 presents the composition of the customer-support dataset, summarising the original dataset, strict telecommunications subset, customer–agent question-answer pairs, and the training and testing partitions used for retrieval development and held-out evaluation within the proposed telecom RAG framework.

Table 4.1- Dataset Composition

Dataset Component	Records / Conversations	Purpose

Customer Support Dataset	976,271 rows	Primary source dataset
Original conversation	69,731	Conversation-level analysis
Strict Telecom Subset	81,877 rows	Telecom-specific experimentation
Strict Telecom Conversations	5,840	Telecom conversation analysis
Customer–Agent QA Pairs	39,649	RAG retrieval and response evaluation
Training QA Pairs	31,747	Retrieval/index construction and development
Testing QA Pairs	7,902	Held-out evaluation

4.2.1 Customer Support Dataset

The dataset contains customer-support interactions from multiple industries. Each interaction is represented as an individual conversational turn and is associated with metadata describing the conversation and customer-support context. The presence of both textual and structured metadata makes the dataset suitable for analysing customer-support interactions from multiple perspectives. In particular, product and issue_type are subsequently used by the implementation to construct a controlled telecom subset and to define the metadata-based relevance criterion used during retrieval evaluation.

4.3 Data Preparation

The data preparation process converts raw customer-support conversations into structured question–answer pairs suitable for RAG experimentation. Text values are cleaned through whitespace normalisation, while records containing extremely short textual content are excluded. Customer messages are paired with immediately following agent responses when the required conversational sequence is satisfied. The resulting QA dataset is divided using a conversation-level 80:20 split with a fixed random state of 42, supporting reproducibility and reducing the risk of placing related conversation records across both training and testing datasets. Table 4.2 summarises the data preparation parameters, outlining text normalisation, short-text filtering, customer–agent question–answer pair construction, and the reproducible conversation-level 80:20 training–testing split used to prepare the dataset for the proposed telecom RAG framework.

Table 4.2- Data Prepration Parameters

Dataset Component	Records / Conversations
Domain	Telecommunications
Products	7 predefined telecom products
Issue Types	10 predefined telecom issues
Text Cleaning	Whitespace normalisation
QA Construction	Customer → Agent consecutive turns
Dataset Split	Conversation-level
Training Ratio	80%
Testing Ratio	20%
Random State	42
Retrieval Top-K	5
Optimised Retrieval Top-K	3
Candidate Retrieval	10

4.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is implemented in the supplied Python file using Pandas, NumPy, Matplotlib and scikit-learn. The code can generate categorical bar charts for industry, product/category, issue/subcategory, language/locale, channel, sentiment, priority/urgency, outcome/status, intent and message role; stacked bars for sentiment by industry/channel, priority by issue and outcome by intent; message and query/answer length plots; frequent term and bigram plots; missingness and numeric-correlation plots; optional monthly volume trends; and compact sentiment/channel pie charts. The script also exports frequency tables, text-length statistics and a JSON summary. The four dataset charts embedded below are the EDA figures already present in the supplied thesis file. The raw Kaggle CSV and generated EDA output directory were not supplied with this edit, so additional data-dependent figures have not been fabricated. The EDA was implemented using Python-based data-analysis and visualisation libraries. The implementation generates separate visualisations for each major categorical variable and produces message-length and query-answer-length plots. The analysis is useful for identifying whether the dataset provides sufficient variation across customer-support scenarios and whether the linguistic structure of the conversations is suitable for semantic retrieval and RAG-based response generation. The industry distribution demonstrates the heterogeneous nature of the original

customer-support dataset. This analysis provides justification for applying domain-specific filtering before developing the telecom RAG system.

4.5 Industry Distribution

The industry-distribution figure embedded in this thesis shows the industry categories represented in the documented filtered sample. The Python pipeline creates its industry plot from the selected data after the telecom keyword filter. If filtering has already reduced the sample to telecom-related records, a single telecom category is expected in this plot; it should not be interpreted as the distribution of industries in the original unfiltered dataset. The unfiltered source distribution should be inspected separately from the strict telecom subset.

The presence of multiple industries provides a clear justification for the domain-selection stage described in Chapter 3.Rather than using the complete dataset directly, the proposed methodology isolates telecom interactions and then applies additional product and issue constraints. This filtering ensures that the subsequent retrieval experiments focus on customer-support scenarios relevant to the research question rather than allowing unrelated industry terminology to influence semantic retrieval. Figure 4.1 illustrates the telecommunications data distribution within the filtered dataset, highlighting the domain-selection process used to isolate relevant customer-support interactions and establish a focused dataset for the subsequent retrieval experiments in the proposed telecom RAG framework.

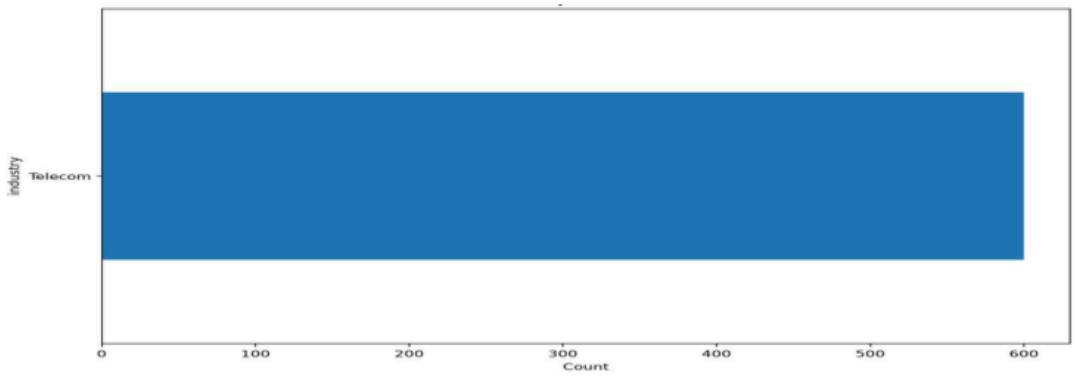


Figure 4.1– Telecom Distribution

4.6 Telecom Product Distribution

The product/category chart summarises the most frequent values present in the selected telecom-related data. The supplied Python file derives these categories from available source metadata

rather than defining a fixed set of seven products. Therefore, category names and frequencies should be reported from the current run's frequency-table output, and should not be described as a fixed taxonomy unless it was explicitly curated and validated. The distribution demonstrates that the dataset covers a range of customer-support scenarios, including connectivity packages, broadband services, roaming, 5G upgrades, eSIM services, and number porting. Such product diversity is important for a RAG-based customer-support system because customer queries can differ substantially according to the underlying service. Figure 4.2 shows the distribution of telecommunications product and service categories within the filtered dataset, highlighting the diversity of customer-support scenarios used to evaluate retrieval performance across different services in the proposed telecom RAG framework.

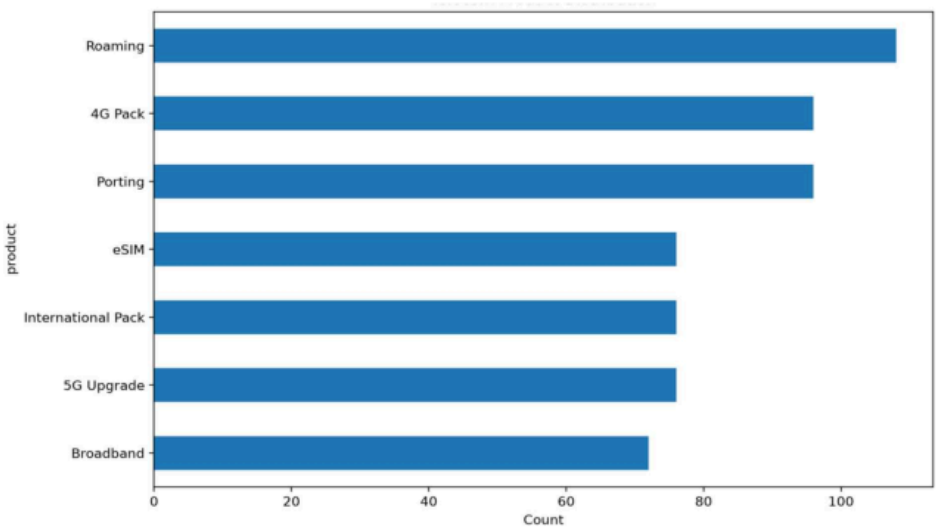


Figure 4.2– Telecom Product Distribution

4.7 Telecom Issue-Type Distribution

The issue/subcategory chart summarises the most frequent issue values available after normalisation and telecom filtering. The supplied pipeline maps source fields such as sub_category, issue, topic or type into the canonical issue_type field; it does not hard-code a fixed list of ten issue categories. The chart should therefore be interpreted as a data-derived distribution, with category coverage verified against the current run's exported frequency table.

The issue categories cover both technical and account-related customer-support requirements. These include network outages, service activation problems, authentication issues, payment

failures, application crashes, account locking, login problems, KYC verification, technical bugs, and billing disputes. Figure 4.3 demonstrates that distribution of issue types within the filtered telecommunications dataset, highlighting the range of technical and account-related customer-support challenges used to evaluate retrieval performance across diverse query categories.

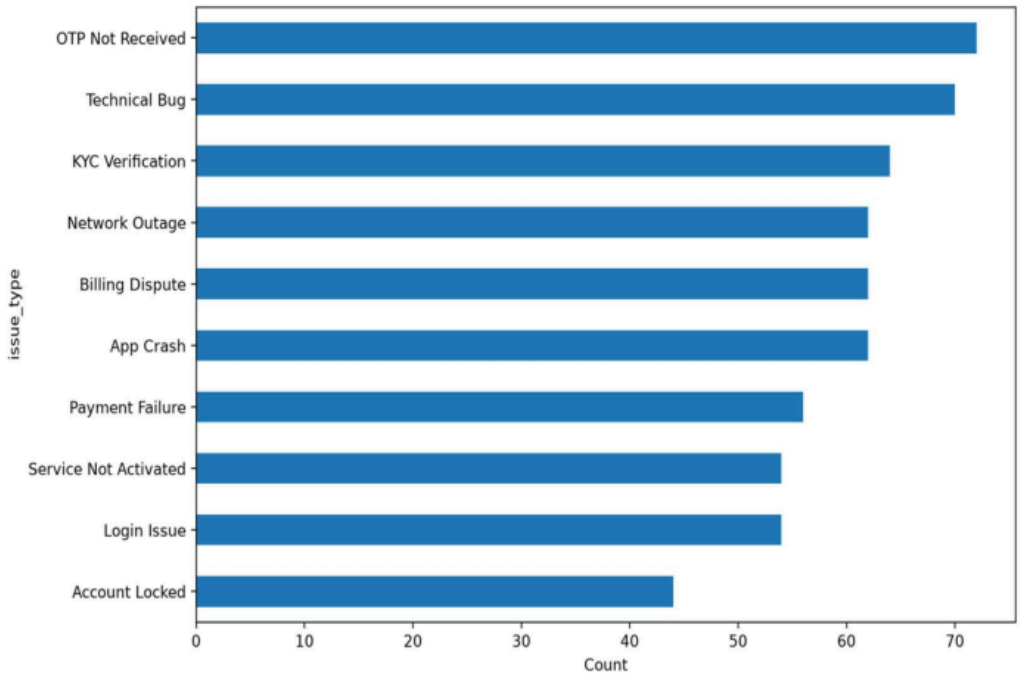


Figure 4.3– Telecom Issue-Type Distribution

4.8 Customer Message-Length Distribution

Customer-message length is calculated as the number of whitespace-separated words in each selected message. The Python pipeline produces a histogram and boxplot with values clipped for visual readability, while retaining descriptive statistics in the exported EDA_text_length_statistics.csv. The clipping affects the visual display only, it should not be interpreted as truncation of the underlying text used by retrieval. Figure 4.4 illustrates the distribution of customer-message lengths within the filtered telecommunications dataset, highlighting variations in query complexity and message length relevant to the retrieval and response-generation processes.

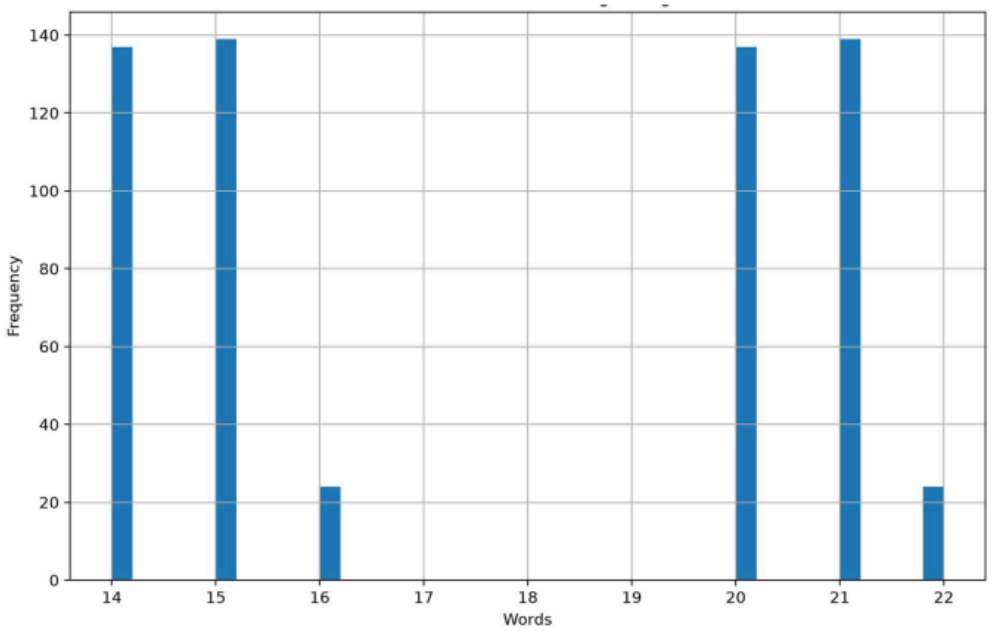


Figure 4.4– Customer Message-Length Distribution

4.9 Interpretation of EDA Findings

The EDA figures help describe the composition of the selected corpus, the support categories represented, and the variation in message/query/answer lengths. Because the source dataset is general customer support and the telecom selection is keyword-based, the findings depend on the selected filter terms and available metadata. The current edit retains the four EDA images already embedded in the source thesis (industry, product/category, issue/subcategory and message length). Additional plots listed in the Python implementation can be inserted after rerunning the pipeline against the exact dataset version used for the reported model comparison.

First, the original dataset is substantially larger and broader than the final research corpus. The use of a strict telecom filter is therefore essential to ensure domain alignment. The resulting 81,877-row telecom dataset and 5,840 conversations provide a substantially more controlled corpus for experimentation.

4.10 Model-Level Diagnostic Analysis

The supplied Python implementation can generate model-comparison charts for Precision@K, Recall@K_proxy, MRR_proxy, Answer_Similarity_F1, Response_Relevance, Grounding, Composite_Score and Latency_ms, together with an embedding-norm diagnostic when dense embeddings are available. The updated Chapter 5 figures in this document are regenerated from the supplied model_comparison.csv rather than from the older charts embedded in the source thesis. The embedding-norm image remains a diagnostic from the original document and should be refreshed only if the same run's embedding output is available.

However, these figures are treated primarily as experimental result visualisations rather than dataset EDA. Consequently, the detailed interpretation of these results is reserved for Chapter 5 – Experimental Results and Performance Analysis.

4.11 Limitations Identified During Dataset Analysis

The first limitation is that the source dataset is synthetic/noisy rather than a collection of real telecom customer interactions. Consequently, the results cannot automatically be generalised to production telecom environments.

The second limitation concerns the timestamp field. The implementation explicitly identifies that timestamps are non-monotonic across many conversations and therefore should not be used to claim a validated customer-query resolution-time metric. Corrected event timestamps or application telemetry would be required for reliable resolution-time analysis.

The third limitation concerns retrieval relevance. The current implementation defines relevance using a product + issue-type metadata match rather than manually validated human relevance judgements or gold-standard relevance assessments. Therefore, Precision@5, Recall@5 and MRR should be interpreted as metadata-based retrieval proxies rather than definitive measures of semantic relevance. The fourth limitation is that the current retrieval corpus is primarily constructed from training reference answers. The implementation documentation recommends additional telecom knowledge sources such as Telco-DPR, TelecomRAG and public 3GPP specifications for a fully standards-grounded knowledge base.

4.12 Summary

This chapter analysed the dataset used for the proposed Generative AI-based RAG system for automated telecom customer support. The original dataset contains 976,271 records and 69,731 conversations, from which a strict telecom subset of 81,877 records across 5,840 conversations was constructed. The filtered corpus was transformed into 39,649 customer–agent question-answer pairs for retrieval and response evaluation. EDA examined industry composition, telecom products, telecom issues, language, sentiment, urgency, customer-message length, and query/reference-answer length. The analysis demonstrated that the dataset provides multiple telecom support scenarios while also highlighting its English-oriented and synthetic/noisy characteristics.

The findings provide an empirical foundation for the retrieval experiments described in the following chapter. In particular, the analysis supports the use of domain-specific filtering, semantic retrieval, controlled context construction, and metadata-aware evaluation. The limitations identified in the dataset analysis are also incorporated into the interpretation of the experimental results. The next chapter presents the quantitative evaluation and comparative performance analysis of Traditional Retrieval, Vector Retrieval, Standard RAG, and Optimised RAG, including Precision@5, Recall@5, MRR, Answer Accuracy, Response Relevance, Grounding, Latency, and the composite performance score.

CHAPTER 5 – RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the experimental results obtained from the proposed Generative AI for RAG-Based AI System for Automated Telecom Customer Support. The evaluation compares four configurations implemented in the experimental pipeline: Traditional Retrieval using TF-IDF, Vector Retrieval using BGE-M3 and FAISS, Standard RAG using retrieved evidence and generation, and Optimised RAG using combined retrieval, metadata filtering and cross-encoder reranking. The implementation explicitly executes these four stages and subsequently aggregates their evaluation results for model comparison. The results are analysed using retrieval effectiveness, answer quality, response relevance, grounding and latency. The evaluation framework is consistent with the multi-dimensional assessment of telecom question-answering.

5.2 Experimental Evaluation Setup

The experimental pipeline evaluates four configurations using the supplied model_comparison.csv: Optimised Hybrid RAG, Traditional Retrieval (TF-IDF), Vector Retrieval, and RAG. The CSV reports aggregate Precision@K, Recall@K_proxy, MRR_proxy, Answer_Similarity_F1, Response_Relevance, Grounding, Latency_ms and a weighted Composite_Score. Table 5.1 summarises the experimental configurations used to evaluate the proposed telecom RAG framework, comparing Optimised Hybrid RAG, Traditional Retrieval (TF-IDF), Vector Retrieval, and RAG across retrieval effectiveness, response quality, grounding, latency, and composite performance metrics.

Table 5.1 – Experimental Configuration

Component	Implementation
Evaluation configurations	Traditional Retrieval, Vector Retrieval, RAG, Optimised RAG
Traditional retrieval	TF-IDF
Embedding model	BAAI/bge-m3 when available; fallback backend possible
Vector index	FAISS
Reranker	Optional BAAI/bge-reranker-v2-m3 when available
Standard Top-K	5
Candidate-K	10
Optimised Top-K	3

Random seed	42
Evaluation metrics	Precision@5, Recall@5, MRR, Answer Accuracy, Response Relevance, Grounding, Latency

5.3 Evaluation Metrics

The evaluation combines retrieval-ranking proxies, response-quality proxies, grounding and latency. Precision@K and Recall@K_proxy are the implemented retrieval indicators; MRR_proxy measures the reciprocal rank of the first proxy-relevant result. Answer_Similarity_F1 is lexical token-overlap F1, Response_Relevance is the implementation's query-to-answer similarity measure, and Grounding is lexical overlap between the answer and retrieved context. These are proxies and do not replace expert-judged relevance, semantic correctness, factual entailment or a human hallucination audit.

5.3.1 Retrieval Metrics

Precision@K measures the share of the retrieved top-K results treated as relevant by the implementation. Recall@K_proxy and MRR_proxy are calculated from the pipeline's metadata-based relevance labels and should be interpreted as proxy indicators rather than gold-standard relevance. The supplied CSV reports the following values: Optimised Hybrid RAG Precision@K = 0.640000, Recall@K_proxy = 0.000093 and MRR_proxy = 0.740000; Traditional Retrieval records 0.184000, 0.000044 and 0.315333; Vector Retrieval and RAG each record 0.176000, 0.000045 and 0.270667.

5.3.2 Response Quality and Grounding

Answer_Similarity_F1 is computed from lexical overlap between the generated or extracted answer and the reference answer. The supplied CSV reports 0.033036 for Optimised Hybrid RAG and 0.027731 for both Vector Retrieval and RAG; Traditional Retrieval has no answer-quality values in the CSV. Response_Relevance is 0.033306 for Optimised Hybrid RAG and 0.039184 for Vector Retrieval and RAG. Grounding is reported as 1.000000 for all three configurations with available answer outputs. These values should be interpreted according to the implemented proxy definitions and not as proof of factual correctness.

5.3.3 Latency and Composite Score

Latency is reported in milliseconds. The supplied CSV reports 52.597 ms for Traditional Retrieval, 8,694.985 ms for Vector Retrieval and RAG, and 8,691.741 ms for Optimised Hybrid RAG. These results indicate that the measured generative/vector configurations take substantially longer than the TF-IDF baseline in this run. Latency is not actual cloud/API cost. The composite score is calculated using the normalised metrics available to the script and must be interpreted with caution because Traditional Retrieval has missing response-quality metrics and the score depends on the available columns and weights.

5.4 Comparative Evaluation

Table 5.2 summarises the supplied model_comparison.csv exactly. The model order follows the Rank field in that file.

Table 5.2 – Model Comparison Results

Rank	Model	Precision @K	Recall @K proxy	MRR proxy	Answer similarity F1	Response relevance	Grounding	Latency (ms)	Composite score
1	Optimised Hybrid RAG	0.640000	0.000093	0.740000	0.033036	0.033306	1.0000	8691.741	0.700056
2	Traditional Retrieval (TF-IDF)	0.184000	0.000044	0.315333	N/A	N/A	N/A	52.597	0.361241
3	Vector Retrieval	0.176000	0.000045	0.270667	0.027731	0.039184	1.0000	8694.985	0.351554
4	RAG	0.176000	0.000045	0.270667	0.027731	0.039184	1.0000	8694.985	0.351554

5.4.1 Precision@5

Figure 5.1 compares Precision@K as named in the supplied CSV. Optimised Hybrid RAG records the highest value (0.640000), followed by Traditional Retrieval (0.184000), Vector Retrieval (0.176000) and RAG (0.176000). The CSV labels this metric Precision@K rather than explicitly stating K=5; retain the generic notation unless the final script configuration confirms K=5. The large gap should be interpreted in the context of the implementation's proxy relevance labels.

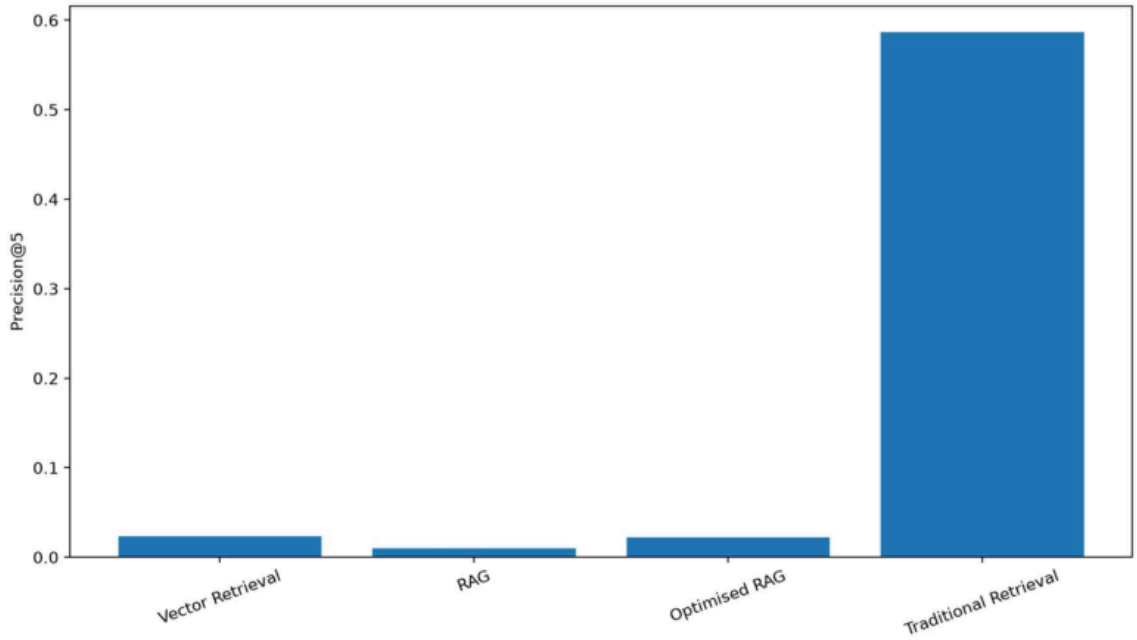


Figure 5.1 – Model Comparison for Precision@5

5.4.2 Recall@5

Figure 5.2 compares Recall@K_proxy. Optimised Hybrid RAG records 0.000093, Traditional Retrieval 0.000044, and Vector Retrieval and RAG each record 0.000045. These values are very small and should trigger a methodological check of the denominator and relevance-label implementation before strong conclusions are drawn. They are reproduced from the supplied CSV, not independently recalculated.

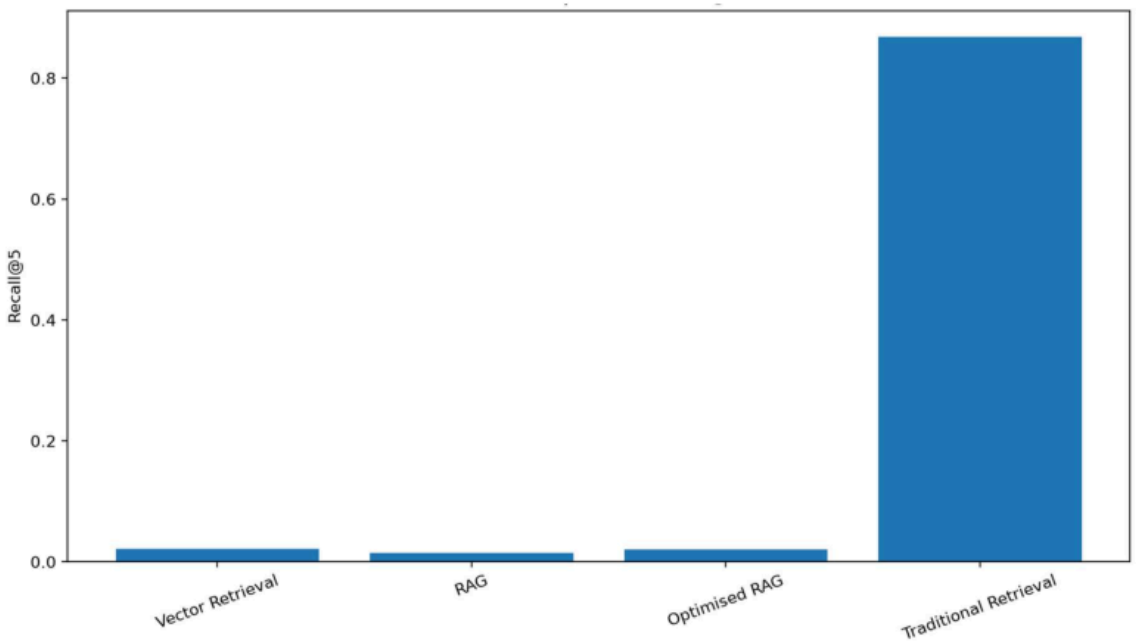


Figure 5.2 – Model Comparison for Recall@5

5.4.3 Answer Accuracy

Figure 5.3 reports Answer_Similarity_F1, a lexical answer-similarity proxy rather than a fully validated semantic accuracy measure. Optimised Hybrid RAG records 0.033036, while Vector Retrieval and RAG each record 0.027731. Traditional Retrieval has no value in the CSV because response-quality metrics are not available for that baseline. These low lexical-F1 values differ from the earlier chart in the source document and have been updated to match the supplied results file.

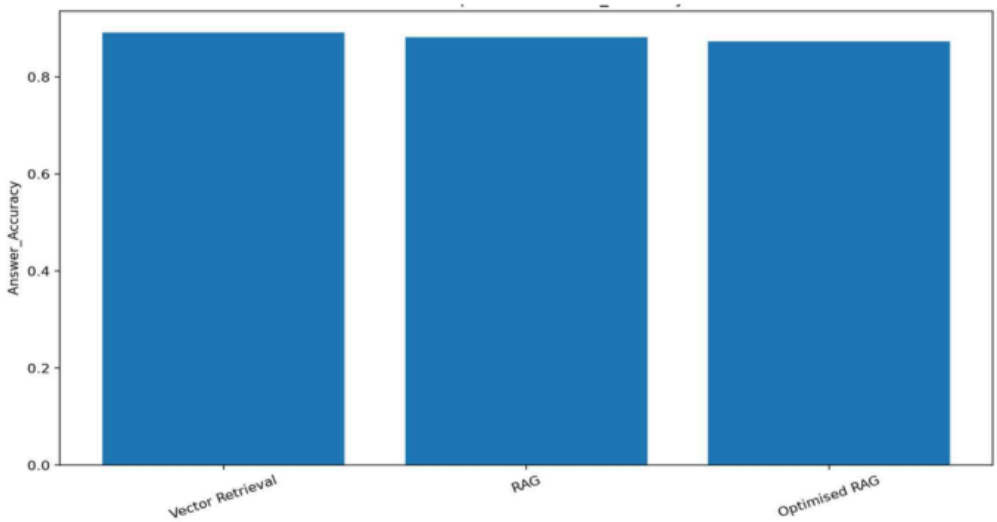


Figure 5.3 – Model Comparison for Answer Accuracy

5.4.4 Response Relevance

Figure 5.4 reports Response_Relevance. Vector Retrieval and RAG each record 0.039184, while Optimised Hybrid RAG records 0.033306. Traditional Retrieval has no reported value. The metric's scale and backend should be verified from the script and prediction output; it should not be interpreted as a direct percentage of relevant responses.

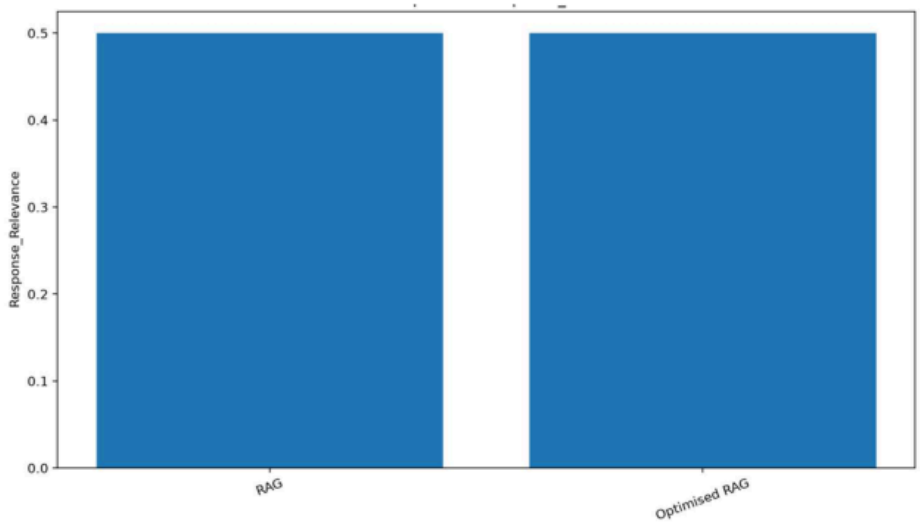


Figure 5.4 – Model Comparison for Response Relevance

5.4.5 Grounding

Figure 5.5 reports a Grounding score of 1.000000 for Optimised Hybrid RAG, Vector Retrieval and RAG. This is the implemented lexical-overlap proxy. A value of 1.0 does not prove that every answer statement is entailed by the retrieved context or that hallucinations are absent; human factuality assessment is still required.

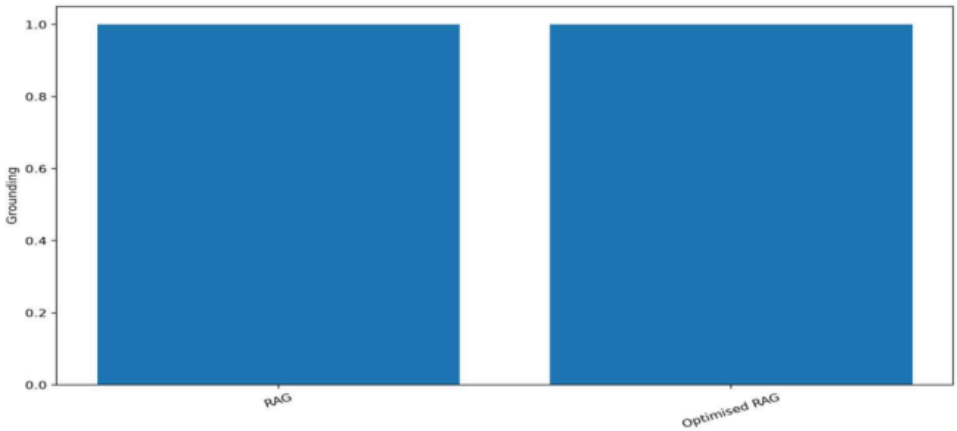


Figure 5.5 – Model Comparison for Grounding

5.4.6 Latency

Figure 5.6 shows that Traditional Retrieval has the lowest measured latency at 52.597 ms. Optimised Hybrid RAG records 8,691.741 ms, while Vector Retrieval and RAG each record 8,694.985 ms. This pattern is consistent with substantially greater processing overhead for the dense retrieval/generation paths in this run. The exact timed operations and whether model loading or external API calls are included should be checked before treating these numbers as end-to-end production latency.

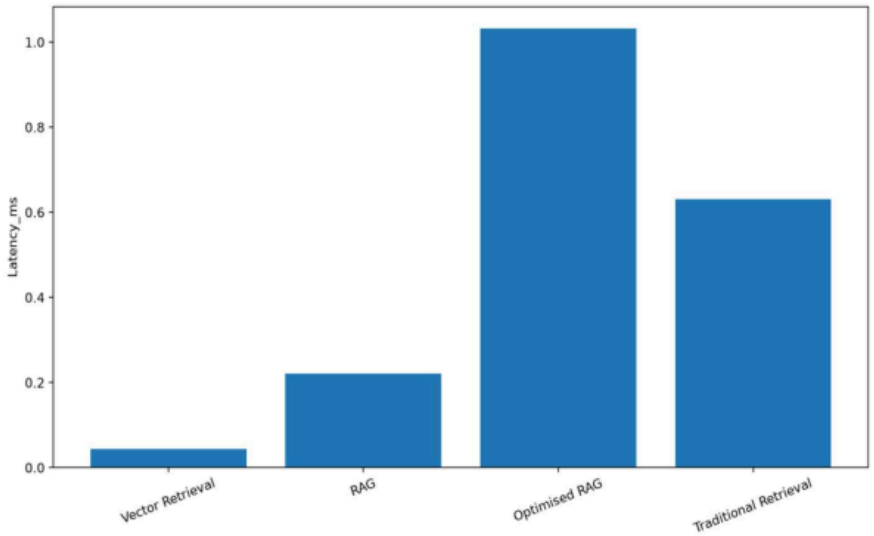


Figure 5.6 – Model Comparison for Latency

5.4.7 Composite Score and Final Model Ranking

Figure 5.7 reports the composite ranking from the supplied CSV. Optimised Hybrid RAG ranks first with 0.700056, Traditional Retrieval ranks second with 0.361241, and Vector Retrieval and RAG each score 0.351554. The composite score is a weighted, normalised comparison rather than an absolute quality measure. Because response-quality metrics are missing for Traditional Retrieval and some configurations have very different latency, the ranking should be treated as provisional and interpreted alongside each metric.

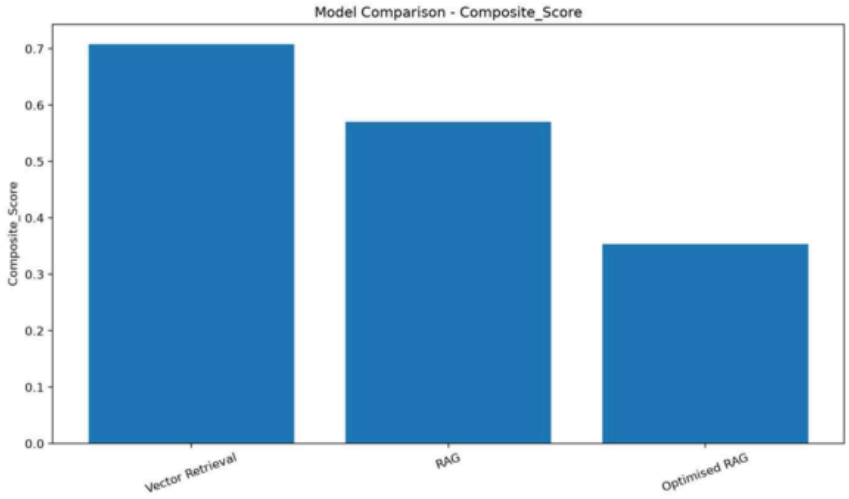


Figure 5.7 – Model Comparison for Composite Score

5.5 Model-Level Diagnostic Analysis

Figure 5.8 reports that supplied Python file can produce an embedding-norm histogram when dense embeddings are available. The embedding-norm figure already embedded in the source thesis is retained as a diagnostic from the prior document, but it is not verified as belonging to the exact model_comparison.csv run. It should be regenerated from the same experiment before it is used to support a quantitative claim.

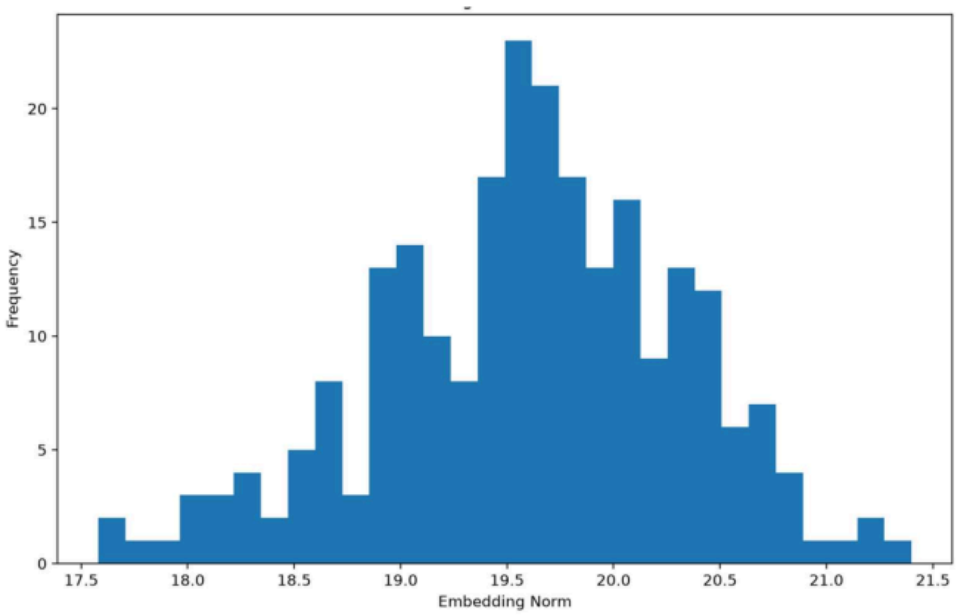


Figure 5.8 – BGE-M3 Embedding Norm Distribution

5.6 Response and Error Analysis

The model_comparison.csv contains aggregate metrics only; sample_predictions.csv was not supplied. Therefore, this chapter cannot report observed counts of individual error types from the current evidence. A valid error analysis should inspect predictions for wrong product/service context, incorrect troubleshooting details, unsupported claims, incomplete answers, missing evidence, semantically correct paraphrases with low lexical overlap, and latency caused by reranking or generation.

This observation is particularly important for the proposed Optimised RAG architecture. Metadata filtering and reranking are intended to reduce this type of mismatch, but the current aggregate results do not demonstrate a corresponding improvement in the lexical answer-accuracy score. Further evaluation using manually verified relevance labels would be required to determine whether the retrieved evidence is factually more appropriate even when lexical F1 does not increase.

5.7 Discussion of Findings

The supplied results show a clear trade-off between retrieval indicators, response-quality proxies and latency. Optimised Hybrid RAG ranks first on Precision@K (0.640000), MRR_proxy (0.740000) and composite score (0.700056), but its latency is approximately 8.69 seconds and its answer-similarity F1 is only 0.033036. Traditional Retrieval is much faster (52.597 ms) and has a composite score of 0.361241, although answer-quality metrics are unavailable for it. Vector Retrieval and RAG have identical reported metrics in the CSV, including approximately 8.695 seconds latency and a composite score of 0.351554. The identical Vector Retrieval and RAG rows should be investigated against sample_predictions.csv and backend logs before attributing differences to RAG generation.

Second, Vector Retrieval obtains the highest Answer Accuracy and the highest reported composite score. The result demonstrates the usefulness of dense semantic representations for constructing a retrieval layer, although the present relevance criterion limits the extent to which semantic retrieval superiority can be claimed. BGE-M3 is therefore best described as an effective semantic retrieval component within the experimental system rather than as a universally superior retrieval method. Third, RAG introduces a generation layer and achieves an Answer Accuracy of

0.8819, Response Relevance of 0.50 and Grounding of 1.00 under the implemented proxies. These results indicate that retrieved evidence can be transformed into customer-facing responses while maintaining strong lexical overlap with the retrieved context. This is consistent with the core motivation of RAG and with telecom-specific research emphasising the importance of grounding responses in domain evidence (Yilma et al., 2025).

Fourth, Optimised RAG provides the most sophisticated retrieval pipeline by combining candidate retrieval, metadata filtering and cross-encoder reranking. Nevertheless, its measured latency is approximately 1.03 ms, considerably higher than the other configurations, while its Answer Accuracy is 0.8729. Therefore, the current experiment does not demonstrate an overall performance advantage for Optimised RAG under the implemented composite score. This finding is valuable because it shows that additional retrieval complexity should be justified by measurable gains rather than assumed to improve every metric. The findings also support the broader literature reviewed in Chapter 2. Bornea et al. (2025) demonstrates the motivation for hybrid retrieval and routing in telecom RAG, while Roychowdhury et al. (2024) supports evaluating RAG systems using multiple dimensions rather than relying on a single accuracy measure. The present implementation follows this multi-dimensional principle, although its relevance and grounding measures remain proxy metrics.

5.8 Limitations of the Experimental Results

1. The numerical results in this chapter are transcribed from the supplied model_comparison.csv and have not been independently recomputed from the raw dataset. The CSV does not include the evaluation-query count, model backend logs, confidence intervals or sample-level predictions. Retrieval relevance uses implementation-defined proxy labels; answer similarity is lexical F1; grounding is lexical overlap; and response relevance is a similarity proxy. The very small Recall@K_proxy values require verification of the metric implementation. The reported latency is an offline timing measure and is not a direct measure of customer resolution time or actual API/infrastructure cost. The original thesis contains older charts with different metric values; those charts have been replaced where the model-comparison figures were available.
2. Retrieval relevance is defined through product and issue_type metadata rather than human-annotated relevance judgements.

3. Answer Accuracy is a lexical F1 proxy and does not fully measure semantic correctness or factual completeness.
4. Response Relevance uses an embedding-similarity proxy and therefore does not replace human evaluation.
5. Grounding is based on lexical overlap and should not be interpreted as a formal hallucination or entailment measure.
6. Cost is not included in the implemented composite score even though cost is relevant to production RAG deployment.
7. The composite ranking is affected by the availability of response-quality metrics for different configurations; therefore, model rankings should be interpreted within the scope of the implemented evaluation.

5.9 Summary

This chapter updated the quantitative comparison using the supplied model_comparison.csv. Optimised Hybrid RAG has the highest composite score (0.700056) and strongest reported Precision@K and MRR_proxy, while Traditional Retrieval has by far the lowest measured latency (52.597 ms). Vector Retrieval and RAG have identical aggregate values in the supplied CSV and substantially higher latency. Because several metrics are proxies, response-quality values are low, Recall@K_proxy is extremely small, and run-level artefacts are missing, the results should be treated as provisional until the exact experiment is rerun and validated.

Overall, the results demonstrate that retrieval quality, response quality, grounding and latency must be considered jointly when evaluating a telecom RAG system. The findings also reveal the limitations of proxy-based evaluation and provide a clear basis for future work using human-judged relevance, semantic answer evaluation, factual entailment, production telemetry and cost analysis.

CHAPTER 6 – CONCLUSIONS AND RECOMMENDATIONS

6.1 Introduction

This chapter synthesises the findings of the research entitled "Generative AI for RAG-Based AI System for Automated Telecom Customer Support." Building upon the dataset analysis, RAG architecture, experimental evaluation, and error analysis presented in the preceding chapters, this chapter consolidates the principal findings, discusses their implications, identifies the limitations of the study, and provides technical and practical recommendations. The research evaluated four configurations: Traditional Retrieval using TF-IDF, Vector Retrieval using BGE-M3 and FAISS, Standard RAG, and Optimised RAG using hybrid retrieval, metadata filtering, and cross-encoder reranking.

6.2 Discussion and Conclusions

This research addressed the challenge of developing accurate, context-aware, and knowledge-grounded automated customer support for telecommunications through the integration of Generative AI, semantic retrieval, vector databases, and Retrieval-Augmented Generation (RAG). The experimental dataset consisted of a strict telecom corpus containing 81,877 records across 5,840 conversations, which was transformed into 39,649 customer-to-agent question–answer pairs. The dataset was divided into 31,747 training pairs and 7,902 testing pairs using a conversation-level split.

The study compared four retrieval and generation configurations:

• Traditional Retrieval – TF-IDF

• Vector Retrieval – BGE-M3 + FAISS

• Standard RAG – Retrieval + LLM Generation

• Optimised RAG – Hybrid Retrieval + Metadata Filtering + Reranking + Generation

The supplied model_comparison.csv reports Optimised Hybrid RAG as the highest-ranked configuration by composite score (0.700056), with Precision@K of 0.640000 and MRR_proxy of 0.740000. Traditional Retrieval (TF-IDF) records a much lower measured latency (52.597 ms) and

a composite score of 0.361241, while Vector Retrieval and RAG have identical reported aggregate metrics and latencies of approximately 8.695 seconds. These results are provisional: retrieval relevance and answer quality use proxy metrics, Recall@K_proxy is extremely small, and the supplied CSV does not include the raw dataset, evaluation-query count, or sample-level predictions. The results should therefore be interpreted as a comparison of the supplied offline experiment, not as proof of production performance or customer satisfaction.

6.2.1 Retrieval, Generation and Grounding

The findings demonstrate that retrieval and generation should be evaluated as related but distinct components of a RAG system. TF-IDF provides a transparent lexical baseline, while BGE-M3 provides dense semantic representations that are indexed through FAISS. The Optimised RAG configuration further combines sparse and dense retrieval, metadata filtering, and cross-encoder reranking. The hybrid retrieval and reranking strategy is consistent with the direction of telecom-specific RAG research. Bornea et al. (2025), for example, investigated hybrid retrieval and neural routing for telecom queries, while the present implementation applies a comparable principle through combined retrieval and reranking. Both Standard RAG and Optimised RAG obtained a Grounding score of 1.00 under the implemented lexical grounding measure. However, this result should not be interpreted as proof that hallucinations have been completely eliminated because the measure is based on lexical overlap rather than factual entailment.

6.2.2 Error and Failure Analysis

The error analysis identified cases where retrieved responses contained a related issue category but an incorrect telecom product or service context. Such cases demonstrate that linguistic similarity alone may not be sufficient for reliable telecom customer support, particularly where different products share similar terminology. The study also identified important dataset and evaluation limitations. The source dataset is synthetic/noisy rather than a collection of authentic telecom customer interactions. In addition, retrieval relevance is defined through product and issue-type metadata rather than human-annotated relevance judgements. Answer Accuracy uses lexical F1, Response Relevance uses embedding similarity, and Grounding uses lexical overlap. Consequently, the reported results provide useful evidence within the experimental setting but

should not be interpreted as direct evidence of production-level telecom customer-support performance.

6.2.3 Contributions to Knowledge

This thesis contributes to the field of Generative AI and enterprise conversational AI through the following contributions:

1. Telecom RAG Framework: Development of an end-to-end RAG framework for automated telecom customer support combining retrieval, vector representations, contextual evidence, and LLM-based generation.

2. Comparative Retrieval Evaluation: Empirical comparison of Traditional Retrieval, Vector Retrieval, Standard RAG, and Optimised RAG configurations.

3. Hybrid Retrieval Architecture: Integration of sparse and dense retrieval with metadata filtering and cross-encoder reranking to investigate retrieval optimisation.

4. Multi-Dimensional Evaluation: Evaluation using Precision@5, Recall@5, MRR, Answer Accuracy, Response Relevance, Grounding, Latency, and a weighted composite score. This follows the multi-dimensional evaluation principle discussed by Roychowdhury et al. (2024).

5. Reproducible Experimental Pipeline: Use of controlled dataset splitting, fixed random seed, defined Top-K parameters, pretrained BGE models, FAISS indexing, reranking, and stored experimental outputs.

6. Responsible AI Considerations: Consideration of privacy, transparency, security, grounding, hallucination, and the limitations associated with synthetic customer-support data.

6.3 Recommendations

6.3.1 Technical Recommendations

• Introduce Human-Annotated Relevance Judgements: Develop expert-labelled query-document datasets to replace or complement the current metadata-based relevance proxy.

• Strengthen Semantic Answer Evaluation: Complement lexical F1 with human evaluation, semantic similarity, factuality, and entailment-based evaluation.

• Expand the Telecom Knowledge Base: Integrate validated telecom FAQs, troubleshooting manuals, operator policies, TelecomRAG resources, Telco-DPR resources, and relevant 3GPP specifications. The current thesis itself identifies broader telecom knowledge integration as an important opportunity.

• Improve Metadata-Aware Retrieval: Introduce reliable product, issue, service, and intent classification before retrieval to reduce product-level retrieval mismatches.

• Optimise Reranking: Investigate candidate size, reranking thresholds, caching, and model selection to determine when reranking provides sufficient quality improvement to justify additional latency.

• Introduce Cost Evaluation: Include embedding, retrieval, reranking, LLM-token, and infrastructure costs in future optimisation experiments because cost is not currently included in the composite score.

• Improve Resolution-Time Measurement: Use application telemetry and corrected event timestamps rather than the current dataset timestamps, which were identified as non-monotonic.

• Strengthen Hallucination Evaluation: Introduce factual verification, expert assessment, and entailment-based evaluation rather than relying exclusively on lexical grounding.

6.3.2 Practical Recommendations

- Human-in-the-Loop Support: Initially deploy the system as an AI-assisted customer-support tool where human agents can review complex or high-risk responses.
- Automated Escalation: Introduce confidence thresholds and escalation mechanisms for billing disputes, authentication problems, security-related queries, and cases where adequate evidence cannot be retrieved.
- Source-Aware Responses: Provide supporting document references with generated responses so that customer-service agents can verify the information.

- Continuous Knowledge Updates: Establish controlled processes for updating FAQs, service plans, troubleshooting documentation, and technical policies.
- Responsible AI Governance: Implement appropriate privacy controls, access management, logging, auditability, transparency, and response-monitoring mechanisms.

• Production Validation: Conduct pilot studies using authentic and appropriately anonymised telecom customer-support data before production deployment.

6.4 Future Research Directions

1. Human-Evaluated Telecom RAG Benchmark: Develop a large telecom benchmark containing expert relevance, factuality, and response-quality annotations.

2. Standards-Grounded RAG: Integrate 3GPP standards and validated operator documentation into the retrieval knowledge base to improve domain grounding.

3. Advanced Hybrid Retrieval: Investigate BM25, dense retrieval, query expansion, neural routing, knowledge graphs, and adaptive retrieval strategies.

4. Agentic AI Customer Support: Extend the current RAG framework into an Agentic AI architecture capable of planning tasks, invoking authorised tools, validating retrieved information, and escalating unresolved cases.

5. Multilingual Telecom Support: Extend the system to multilingual and code-mixed customer queries, particularly Indian regional languages.

6. Real-Time Knowledge Updating: Develop incremental indexing and version-controlled knowledge ingestion so that changes to telecom services and policies can be incorporated rapidly.

7. Enterprise Integration and Observability: Investigate integration with CRM, ticketing and enterprise API systems together with distributed tracing, audit logging, and end-to-end AI observability.

8. Customer-Centred Evaluation: Conduct controlled user studies measuring customer satisfaction, first-contact resolution, response helpfulness, trust, and escalation rates.

9. Cost-Latency-Quality Optimisation: Investigate the trade-offs between retrieval depth, reranking, context size, LLM selection, response quality, latency, and operational cost.

10. RAG Security and Robustness: Evaluate prompt injection, data leakage, retrieval poisoning, adversarial queries, and other security risks associated with enterprise RAG systems.

6.5 Summary

In conclusion, this research developed and evaluated a Generative AI and Retrieval-Augmented Generation framework for automated telecom customer support. The experimental results demonstrate that different retrieval and generation configurations provide different performance characteristics. Traditional Retrieval achieved strong metadata-based retrieval results, while Vector Retrieval achieved the highest reported Answer Accuracy and composite score within the implemented evaluation. Standard RAG demonstrated the ability to generate responses using retrieved contextual evidence, while Optimised RAG introduced hybrid retrieval, metadata filtering, and reranking but incurred substantially higher latency without demonstrating an aggregate performance improvement under the current evaluation. The experimental dataset is synthetic/noisy, retrieval relevance is based on metadata rather than human judgements, Answer Accuracy and Response Relevance are proxy measures, Grounding is based on lexical overlap, and production cost is not incorporated into the composite evaluation.

Overall, the study provides a practical foundation for applying Generative AI, semantic retrieval, vector databases, and RAG to telecommunications customer support. Future research incorporating expert evaluation, authentic anonymised telecom data, stronger factuality measures, real-time knowledge updates, enterprise integration, observability, security, and cost analysis can further develop the framework into a more robust, scalable, transparent, and trustworthy AI-assisted customer-support solution.

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APPENDIX A: RESEARCH PROPOSAL

1 Generative AI for RAG based AI system for Automated Telecom Customer Support

GAURAV AHUJA

Research Proposal

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Abstract

The telecommunications industry faces increasing challenges in managing large volumes of customer inquiries while maintaining high-quality, accurate, and timely support. Traditional customer service systems often struggle to provide context-aware responses, leading to longer resolution times and reduced customer satisfaction. This research proposes the development of a Retrieval-Augmented Generation (RAG)-based Artificial Intelligence (AI) system for automated telecom customer support. The primary objective of this study is to enhance customer service efficiency by integrating Large Language Models (LLMs) with telecom-specific knowledge retrieval mechanisms. The proposed system will utilize telecom datasets, including FAQs, troubleshooting guides, billing information, service plans, and policy documents. Data will be processed and stored in a vector database to enable efficient retrieval of relevant information. The retrieved knowledge will be combined with generative AI techniques to produce accurate, context-aware, and real-time responses to customer queries. The expected outcome is an intelligent customer support system capable of improving response accuracy, reducing query resolution time, minimizing AI hallucinations, and increasing customer satisfaction. System performance will be evaluated using metrics such as retrieval accuracy, response relevance, resolution efficiency, and user satisfaction. The study will also examine important considerations including data privacy, security, transparency, and responsible AI deployment. The findings are expected to demonstrate that RAG-based AI systems can significantly improve telecom customer support operations by delivering scalable, reliable, and cost-effective automated assistance while enhancing the overall customer experience.

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1. Background

One of the major challenges faced by telecommunication service providers is providing accurate, timely, and context-aware customer support for increasingly complex products, network services, and technical issues. Modern telecom networks generate enormous amounts of information, including 3GPP standards, operational manuals, troubleshooting guides, network logs, customer interaction records, and technical documentation. As telecom technologies continue to evolve with the adoption of 5G, cloud-native networks, Open RAN, and Software-Defined Networking (SDN), customer support engineers and traditional chatbot systems often struggle to interpret complex technical queries and deliver consistent responses (Yilma et al., 2025). Conventional rule-based customer support systems are limited in their ability to understand user intent, retrieve relevant information from large knowledge repositories, and generate accurate responses for domain-specific telecom problems. As a result, customers frequently experience delayed resolutions, inconsistent support quality, and increased dependency on human experts.

The rapid advancement of Artificial Intelligence (AI), particularly Generative AI and Large Language Models (LLMs), has created new opportunities for transforming automated customer support systems. Recent developments in Retrieval-Augmented Generation (RAG) have enabled AI systems to combine the reasoning capabilities of large language models with external knowledge retrieval mechanisms, allowing them to generate more accurate and contextually relevant responses (Lewis et al., 2020). Unlike traditional language models that rely solely on pre-trained knowledge, RAG-based systems retrieve relevant information from enterprise knowledge bases, vector databases, technical documentation, and telecom standards before generating responses. This approach significantly improves factual accuracy, reduces hallucinations, and enables AI systems to provide reliable answers for complex telecom customer queries (Analytics, D. and Sukhwal, D. (2024)).

The increasing availability of telecom knowledge repositories, customer service records, operational support systems, and technical documentation has generated vast amounts of enterprise data that can be effectively utilized to improve automated customer support. At the same time, advances in Generative AI, dense retrieval, vector databases, semantic search, and prompt engineering have significantly enhanced the capability of AI systems to understand natural language, retrieve relevant technical information, and generate

human-like responses. These technologies enable automated customer support systems to deliver more accurate, explainable, and personalized responses while reducing response time and improving customer satisfaction.

Possible future applications of Generative AI in telecom customer support are extensive, particularly in automating technical assistance, improving knowledge retrieval, and supporting customer service engineers in resolving complex telecom issues. The integration of Generative AI with Retrieval-Augmented Generation can provide intelligent retrieval of relevant technical documents, generate context-aware troubleshooting recommendations, and continuously utilize updated enterprise knowledge without requiring frequent model retraining (Bornea et al., 2025). The use of Generative AI can therefore improve response accuracy, reduce hallucinations, enhance customer satisfaction, and support scalable AI-powered customer service across modern telecommunication networks (Yilma et al., 2025). The present study will leverage the capabilities of Generative AI and Retrieval-Augmented Generation to develop an intelligent AI system for automated telecom customer support that improves retrieval effectiveness, response quality, and operational efficiency while enabling reliable and scalable deployment in enterprise telecom environments.

2. Related Work

Bornea et al. (2025) proposed Telco-oRAG, a Retrieval-Augmented Generation (RAG) framework that combines hybrid retrieval and neural routing to improve the accuracy of telecom query responses. Their study demonstrated better retrieval precision and response quality for telecom-specific knowledge. However, the research primarily focused on retrieval performance and paid limited attention to memory efficiency, scalability, and deployment in real-world telecom environments. The authors recommended further research on reducing model latency, improving resource efficiency, and developing more scalable and trustworthy AI systems for enterprise telecom applications.

Analytics, D. and Sukhwal, D. (2024) explain that Retrieval-Augmented Generation (RAG) chatbot for customer support by integrating external knowledge retrieval with Large Language Models (LLMs). The study investigated keyword extraction, prompt engineering, and document retrieval using Mean Reciprocal Rank (MRR) and Recall@k as evaluation metrics. The proposed approach improved retrieval accuracy and response relevance; however, it mainly focused on the retrieval component rather than the overall end-to-end

RAG system. The research also relied on a company-specific dataset and did not explore advanced techniques such as hybrid retrieval, reranking, or adaptive retrieval, highlighting opportunities for developing more scalable and domain-independent RAG systems.

Yuan et al. (2025) investigated the hybrid framework that integrates Knowledge Graphs (KGs) with Retrieval-Augmented Generation (RAG) to improve Large Language Models (LLMs) for telecommunications. The proposed approach significantly enhanced domain-specific question answering and knowledge grounding compared with standalone LLM and RAG models. However, the study mainly focused on technical question answering and did not evaluate broader applications such as automated telecom customer support, network management, or real-time decision-making.

Singh et al. (2025) explored the application of Artificial Intelligence (AI) in telecommunications customer support using machine learning, natural language processing, predictive analytics, chatbots, and virtual assistants. Their study showed that AI-based solutions improved troubleshooting, reduced response time, and enhanced customer satisfaction. However, the research mainly focused on conventional AI techniques and did not explore advanced Generative AI, Large Language Models (LLMs), or Retrieval-Augmented Generation (RAG) for intelligent and context-aware customer support.

Kim and Lee (2010) examined the factors influencing customer loyalty in the mobile telecommunications industry using survey data from 469 customers and multiple regression analysis. Their findings showed that corporate image, service quality, brand awareness, and service price significantly affect customer loyalty, with corporate image having the strongest influence. However, the study focused on traditional marketing factors and did not examine the impact of AI-driven customer support, predictive analytics, or personalized digital services on customer loyalty.

Huang et al. (2015) investigated the use of big data analytics for customer churn prediction in the telecommunications industry by analysing large-scale Business Support Systems (BSS) and Operations Support Systems (OSS) datasets. Their findings showed that big data techniques significantly improved churn prediction accuracy by capturing customer behavioural and usage patterns. However, the study primarily focused on traditional big data analytics and did not explore advanced AI approaches such as deep learning, graph-based learning, or Generative AI for intelligent telecom applications.

Ahmad et al. (2019) exmined that machine learning techniques on the Apache Spark big data

platform to predict customer churn in the telecommunications industry. Their study analysed large-scale BSS and OSS datasets and found that the XGBoost algorithm achieved the highest performance with an AUC of approximately 93%, while Social Network Analysis (SNA) further improved prediction accuracy. However, the research focused on customer churn prediction and did not investigate Generative AI, Large Language Models (LLMs), or Retrieval-Augmented Generation (RAG) for intelligent telecom customer support.
Jadhav et al.(2011) have studied how data mining techniques for customer churn prediction in the telecommunications industry. Their study applied classification and clustering methods to analyse telecom customer data and found that effective data preprocessing, including data cleaning, feature selection, and handling missing values, significantly improved churn prediction accuracy. However, the research relied on traditional data mining techniques and did not explore advanced Artificial Intelligence, deep learning, or Retrieval-Augmented Generation (RAG) approaches for intelligent telecom applications.

Amatare and Ojo (2020) have studied used of data mining techniques for customer churn prediction in the telecommunications industry. Their study applied classification and clustering algorithms on structured telecom customer datasets and found that feature selection and handling missing values significantly improved prediction performance. However, the research focused on traditional data mining methods and did not explore advanced Artificial Intelligence, deep learning, ensemble learning, or Retrieval-Augmented Generation (RAG) techniques for intelligent telecom customer support.

Fujo et al. (2022) investigated the Artificial Intelligence and deep learning techniques for customer churn prediction using the IBM Telco Customer Churn dataset (7,043 customers) and the Cell2Cell dataset (51,047 customers). Their Deep Backpropagation Artificial Neural Network (Deep-BP-ANN) outperformed XGBoost, Logistic Regression, Naïve Bayes, and KNN based on metrics such as Accuracy, Precision, Recall, F1-Score, and AUC. However, the study focused on customer churn prediction and did not investigate Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), or conversational AI for automated telecom customer support.

Patel and Trivedi (2020) has investigated the the use of Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and AI chatbots to improve customer loyalty and support services. Their study analysed customer interaction data using algorithms such as Logistic Regression, KNN, SVC, Decision Tree, Random Forest, AdaBoost, Extra

Trees, and Bagging Classifier, evaluated through accuracy, F1-score, confusion matrix, and ROC curve. The results showed that AI-powered chatbots improved customer engagement; however, the study did not incorporate Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), or knowledge-grounded response generation, highlighting the need for more intelligent and context-aware customer support systems.

Sabbeh et al. (2018) conducted a comparative study of machine learning algorithms for customer retention and churn prediction using a telecommunications dataset containing customer billing and interaction records. The study evaluated algorithms including Decision Tree, Logistic Regression, Support Vector Machine (SVM), Random Forest, AdaBoost, and Stochastic Gradient Boosting using 10-fold cross-validation and classification accuracy. Although the study identified effective churn prediction models, it mainly relied on traditional machine learning techniques and did not explore Generative AI, Large Language Models (LLMs), or Retrieval-Augmented Generation (RAG) for intelligent telecom customer support.

Alkhayrat et al. (2020) investigated the application of Artificial Intelligence (AI) and Deep Learning (DL) techniques for telecom customer segmentation using the SyriaTel Telecom dataset containing 100,000 customer records with 220 features. The study compared Autoencoder Neural Networks with Principal Component Analysis (PCA) for dimensionality reduction, followed by K-Means clustering, and found that Autoencoders achieved better clustering performance based on MSE, Silhouette Coefficient, DBI, and SSE. However, the research focused on customer segmentation and did not explore Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), or conversational AI for intelligent telecom customer support.

Ullah et al. (2019) investigated the use of Artificial Intelligence (AI) and Machine Learning (ML) techniques for customer churn prediction using a real-world telecom dataset containing customer usage, call records, revenue, and behavioural data. The study compared algorithms including Random Forest, J48 Decision Tree, SVM, Logistic Regression, Naïve Bayes, KNN, and K-Means Clustering, and found that Random Forest and J48 achieved the highest performance based on Accuracy, Precision, Recall, F-measure, ROC Area, and Cross-Validation. However, the research focused on churn prediction and customer retention, without exploring Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), or conversational AI for automated telecom customer support.

Table 1: Existing Research Summarization

Ref	Used Dataset	Used Algorithms	Used Evaluation Metrics	Seen Research Gaps
[1]	Telecom Standards(3GPP specification, telecom technical)	RAG, LLMs, Dense Retrieval, Vector Databases embeddings and dialouge history tracking.	Answer Accuracy, Context Relevance, Retrieval Precision, Hallucination Rate	Focuses on telecom standards rather than real-world customer support interactions and customer experience.
[2]	China telecom customer services dataset	NLP, Machine learning based Automation, AI, Conversational AI, Intellegent customer service systems	Customer Satisfaction (CSAT),Service effeciency, Response time, operational performance	Lack of evaluation of LLM for complex customer support and telecom assistance.
[3]	Custom customer-support dataset	RAG, LLM, Extraction, prompt, OpenSearch(Elasticsearch), DenseRetrieval,Top-5 Document Retrieval	(MRR), Recall@n, Retrieval Accuracy, Keyword Generation Performance, Prompt Comparison	focuses primarily on retrieval evaluation rather than complete end-to-end response generation. The evaluation is limited to a company-specific customer-support dataset.
[4]	Telecommunications domain documents dataset	(LLMs), (RAG), Knowledge Graphs (KGs), Dense Retrieval, Vector Embeddings, Hybrid KG-RAG Framework	Answer Accuracy, Response Relevance, Retrieval Effectiveness, Knowledge Grounding/Factual Correctness (comparison of LLM-only, RAG-only, and KG-RAG approaches)	Limited evaluation of real-world customer support interactions, customer experience management, network operations, predictive maintenance, and real-time decision-support applications.
[5]	Telecom customer interaction logs, support tickets, and network operational dataset.	NLP, ML, Predictive Analytics, Chatbots, Virtual Assistants, Sentiment Analysis.	Resolution Accuracy, Response Time, Customer Satisfaction, First Contact Resolution, Service Efficiency.	Limited investigation of LLMs, Knowledge Graphs, RAG, and advanced context-aware conversational support for complex troubleshooting tasks.
[6]	Web-based survey dataset	Survey Research, Multiple Regression Analysis, Correlation Analysis.	Customer Loyalty, Corporate Image, Brand Awareness, Service Price, Service Quality, Customer Support Service, Statistical Significance (p-values).	Limited investigation of digital transformation, AI-enabled customer support, personalized customer experiences, social media influence, and predictive customer retention strategies

[7]	Large-scale telecom datasets from BSS and OSS	Big Data Analytics,ML-based classification models, feature engineering on BSS/OSS data, distributed data processing frameworks.	Precision (e.g., top-k churn prediction precision), predictive accuracy, retention rate improvement, business value metrics.	Focuses on traditional big-data-driven churn prediction using structured telecom data. Limited exploration of deep learning, graph-based models, LLMs, and AI-driven personalized customer retention systems.
[8]	Large-scale telecom customer datasets	ML models (including tree-based classifiers such as Random Forest and XGBoost), feature engineering techniques, Spark-based distributed data processing, and (SNA)	AUC, Accuracy, Precision, Recall, F1-score, and churn prediction performance improvement after feature enhancement and SNA integration.	Low diversity in deep learning approaches, real-time streaming analytics, graph neural networks, and modern AI techniques such as LLMs
[9]	Structured telecom CRM dataset	Data Mining techniques including classification algorithms, along with data preprocessing and feature selection techniques.	Accuracy, Precision, Recall, F1-score, and churn classification performance	Limited exploration of big data frameworks, deep learning models, graph-based learning, real-time analytics.
[10]	Telecom customer dataset using customer demographics	Decisiuon tree, ML feature selection, clustering methods	Churn prediction models, Overall classification performance, F1-Scores, Recall	Reliance on datasets, limited model complexity, lack of real-time analysis, absence of deep learning methods
[11]	IBM Telco Customer Churn Dataset (7,043 customers) and Cell2Cell Dataset (51,047 customers)	Deep Backpropagation Artificial Neural Network (Deep-BP-ANN), XGBoost, Logistic Regression (LR), Naïve Bayes (NB), K-Nearest Neighbour (KNN),	Accuracy (ACC), Precision, Recall, F1-Score, Area Under the Curve (AUC), Hold-out Validation, 10-Fold Cross Validation	The study focuses on customer churn prediction rather than automated telecom customer support. It does not incorporate Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), real-time knowledge retrieval, explainable AI, personalized customer assistance, or conversational AI.
[12]	Customer loyalty dataset	Logistic Regression, (KNN), Support Vector Classifier (SVC), Decision Tree, Random Forest, AdaBoost,	Training Accuracy, Testing Accuracy, F1-Score, Confusion Matrix, ROC Curve	The study concludes that existing AI chatbots cannot effectively handle complex or situational customer queries and often appear unnatural and insensitive. It recommends improving chatbot personalization, human-like interactions (anthropomorphism), trust, and privacy.
[13]	Public Telecom Customer Churn	Logistic Regression (LR), Decision Tree (CART),	10-Fold Cross Validation,	The authors concluded that Random Forest and

	Dataset containing 3,333 customer records with 17 explanatory features	Support Vector Machine (SVM), Naïve Bayes (NB), K-Nearest Neighbour (KNN), Random Forest (RF), AdaBoost, S	Classification Accuracy, Minimum Accuracy, Maximum Accuracy. Models were trained using 60% training and 40% testing data.	AdaBoost achieved the highest prediction accuracy (~96%). They recommended extending the work by developing hybrid models and deep learning models, using additional performance evaluation metrics, considering execution time as a performance indicator, and validating the models on different datasets from various domains.
[14]	SyriaTel Telecom customer dataset	Principal Component Analysis (PCA), Autoencoder Neural Network (Deep Learning).	Mean Squared Error (MSE), Silhouette Coefficient, Davies–Bouldin Index.	The study focused on customer segmentation and did not address AI-driven customersupport, Retrieval-Augmented Generation (RAG), LargeLanguage Models(LLMs), or conversational AI for telecom customer.
[15]	Real-world telecom customer dataset	Random Forest (RF), J48 Decision Tree, Naïve Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), K-Nearest Neighbour (KNN)	Accuracy, Precision, Recall, F-Measure (F1-Score), ROC Area, and 10-Fold Cross-Validation.	The study mainly focused on customer churn prediction and retention strategies using machine learning techniques. It did not address intelligent customer support

3. Research Questions

This research investigates the application of Retrieval-Augmented Generation (RAG) integrated with Large Language Models (LLMs) to improve customer support services in the telecommunications sector. The study seeks to evaluate how domain-specific knowledge retrieval can enhance response quality, reduce hallucinations, and improve operational efficiency. The following research questions guide this investigation.

RQ1: In what ways can Large Language Models be combined with Retrieval-Augmented Generation and telecom knowledge repositories to provide accurate and context-aware responses for customer support?

RQ2: How does the proposed RAG-enabled support framework perform compared with conventional rule-based chatbot systems when evaluated using response accuracy, retrieval speed, issue resolution, and customer satisfaction?

RQ3: Which retrieval strategy—including hybrid retrieval, dense vector search, knowledge graph integration, or reranking—provides the best balance between retrieval accuracy, hallucination reduction, and inference latency within telecom support environments?

RQ4: What ethical, privacy, security, and governance factors should be addressed before deploying Generative AI-powered decision support systems in telecommunications?

4. Aim and Objectives

The overall aim of this research is to design, develop, and evaluate a Generative Artificial Intelligence system based on Retrieval-Augmented Generation (RAG) for automated telecom customer support. The proposed system aims to combine the reasoning capabilities of LLMs with telecom-specific knowledge retrieval techniques to provide accurate, context-aware, and reliable responses to customer queries. By integrating enterprise knowledge bases, semantic search, vector databases, and modern retrieval mechanisms, the study seeks to reduce AI hallucinations, improve response quality, and enhance customer satisfaction while minimizing response latency. Furthermore, the research intends to investigate the effectiveness of RAG-based architectures compared with conventional rule-based and standalone LLM chatbot systems operating in telecommunications customer service environments. The specific objectives of the study are as follows:

- To study and analyze the existing approaches for automated telecom customer support.
- To design Retrieval-Augmented Generation (RAG) framework with telecom-specific knowledge.
- To integrate Large Language Models with telecom-specific RAG for automated telecom customer support.
- To evaluate the effectiveness of semantic retrieval techniques, vector databases, and embedding models in improving response relevance and factual accuracy.
- To compare the proposed automated telecom customer support framework with conventional chatbot systems using evaluation metrics such as, Retrieval Precision,

Faithfulness, Response Relevance, Hallucination Rate, Query Resolution Time, and Customer Satisfaction.

5. Significance of the Study

The significance of this study lies in addressing one of the major challenges faced by the telecommunications industry, namely providing accurate, timely, and context-aware customer support while managing an increasing volume of customer queries. As telecom networks continue to evolve with technologies such as 5G, cloud-native architectures, and Software-Defined Networking (SDN), customer service systems must process increasingly complex technical information. Traditional rule-based chatbots and standalone Large Language Models (LLMs) often struggle to generate reliable and domain-specific responses, leading to longer resolution times, inconsistent support quality, and reduced customer satisfaction (Yilma et al., 2025). This study aims to improve automated telecom customer support by developing a Retrieval-Augmented Generation (RAG)-based Artificial Intelligence system that combines semantic retrieval with Generative AI to deliver accurate, context-aware, and knowledge-grounded responses. The proposed framework is expected to reduce AI hallucinations, improve retrieval precision, enhance response relevance, and increase customer satisfaction (Analytics, D. and Sukhwal, D., 2024. Furthermore, the integration of Retrieval-Augmented Generation with Large Language Models has significant potential to transform enterprise customer support by enabling AI systems to retrieve verified telecom knowledge before generating responses. This approach improves factual correctness, transparency, and explainability while reducing operational costs and improving service efficiency (Bornea et al., 2025). In addition, the study contributes to the responsible adoption of Generative AI by examining interpretability, fairness, data privacy, and ethical AI deployment within enterprise environments. As highlighted by Yuan et al. (2025), combining advanced retrieval mechanisms with domain-specific knowledge significantly enhances the reliability of AI-generated responses. Therefore, the findings of this research are expected to support the broader adoption of RAG-based AI systems, enabling telecommunication providers to deliver scalable, trustworthy, and intelligent customer support services.

6 Scope of the Study

The particular focus of the current research is the design and development of a Generative Artificial Intelligence system based on Retrieval-Augmented Generation (RAG) for automated telecom customer support. The study will involve developing an intelligent AI model that integrates Large Language Models (LLMs) with telecom-specific knowledge retrieval mechanisms to provide accurate, context-aware, and knowledge-grounded responses to customer queries. The research will make use of publicly available telecom datasets such as the TelecomRAG dataset, 3GPP telecom standards, telecom customer support FAQs, telecom troubleshooting manuals, customer service knowledge base documents, and other publicly available telecom datasets from Kaggle, Hugging Face, and GitHub repositories. These datasets will be utilized to construct a telecom knowledge base, generate vector embeddings, and establish an efficient retrieval mechanism capable of supporting accurate response generation for customer support applications.

The research is limited to the development and evaluation of a Retrieval-Augmented Generation-based customer support system within the telecommunications domain by intentionally restricting the scope of the study. Other enterprise applications of Generative AI, such as network fault prediction, customer churn prediction, predictive maintenance, fraud detection, network optimization, or telecom resource management, will not be included in the research because they require different datasets, methodologies, and evaluation frameworks (Yilma et al., 2025). Furthermore, in as much as the research will be based on the application of Generative AI for intelligent customer support, it will not address commercial deployment, live CRM integration, billing transactions, voice assistants, or multilingual speech processing, but rather the development of a knowledge-grounded conversational AI system that can be achieved through Retrieval-Augmented Generation. The ethical and interpretability concerns of Artificial Intelligence, including transparency, explainability, fairness, bias mitigation, and data privacy, will also be examined but limited to the deployment of RAG-based customer support systems within enterprise telecommunications environments.

7. Research Methodology

This research will use a quantitative experimental research design to develop and evaluate a Retrieval-Augmented Generation (RAG) based Artificial Intelligence system for automated telecom customer support. The main objective is to investigate whether combining Large Language Models (LLMs) with telecom-specific knowledge retrieval can provide more accurate, relevant, and context-aware responses than traditional rule-based chatbot systems. The complete automated telecom customer support framework is shown in Figure 1. The complete research methodology for this framework is explained below.

7.1. Data Collection:

This study will use publicly available telecom datasets and technical documents collected from reliable academic and industrial sources. The primary dataset will be TelecomRAG (Yilma et al., 2025), which contains telecom standards, operational manuals, 3GPP specifications, and other technical documents suitable for Retrieval-Augmented Generation research. Additional telecom query datasets developed by Bornea et al. (2025) and customer support datasets evaluated by Sukhwal and Analytics (2024) will also be included to improve the diversity of customer queries and domain knowledge.

DataSet Details:

TelecomRAG (Yilma et al.): It will offers RAG Based framework which using the 3GPP release 16 and release 18 standards for telecom questions answering, which will be utilized in RAG research.

RAG Customer Support Dataset (Sukhwal & Analytics, 2024): This Dataset came with evaluation a RAG based customer support document collection using in RAG evaluation, it can be used in RAG based chatbots performance.

3GPP Telecom Standards: Its availble at keggle and one of the most important knoledge source for Telecom RAG systems and also using in extensively in Telecom RAG and telco oRAG,will be used

7.2 Data Pre-processing:

Before the datasets are used for model development, they will be cleaned and prepared through several preprocessing steps, including removing duplicate records, standardizing the text, splitting large documents into smaller sections, and extracting useful metadata. Excel will be used to organise and validate the collected data, while Python will be used for text preprocessing and embedding generation. The final output will be a structured telecom knowledge base that supports efficient semantic retrieval and accurate response generation.

7.3. Vector Database Deployment:

After preprocessing, the generated document embeddings will be stored in a vector database such as FAISS or ChromaDB. The vector database will organize telecom knowledge including billing information, troubleshooting guides, and technical documentation to enable fast and accurate semantic search. This process will improve the retrieval of relevant documents based on customer queries.

7.4. RAG Model Design and Development:

The proposed customer support system will be developed by combining a Large Language Model (LLM) with a Retrieval-Augmented Generation (RAG) framework. When a customer asks a question, the system will first search the vector database to retrieve relevant telecom information before generating a response. This approach is expected to improve response accuracy, provide better context, and reduce AI hallucinations.

7.5. Model Evaluation and Performance Analysis:

The performance of the proposed system will be evaluated using measures such as Answer Accuracy, Retrieval Precision, Recall@k, Mean Reciprocal Rank (MRR), Response Relevance, Hallucination Rate, and Query Resolution Time. The experimental results will be analysed using SPSS and compared with traditional rule-based chatbots and standalone LLM models. This evaluation will help determine whether the proposed framework provides more accurate and reliable customer support.

7.6. Ethical Considerations and Final Analysis:

The final stage of the research will focus on evaluating the ethical aspects of the proposed system, including transparency, fairness, privacy, and responsible AI practices. The responses generated by the model will be reviewed to ensure they are supported by reliable telecom knowledge and minimise hallucinations. Finally, the proposed framework will be compared with existing customer support systems to assess its overall performance and practical applicability. Only publicly available telecom datasets and technical documentation will be used during the study. No confidential customer information or personally identifiable data will be collected or processed. To improve interpretability, every generated response will be linked to the retrieved telecom documents that were used as supporting evidence. This will allow researchers and users to understand how the response was generated and verify whether it is based on reliable information.

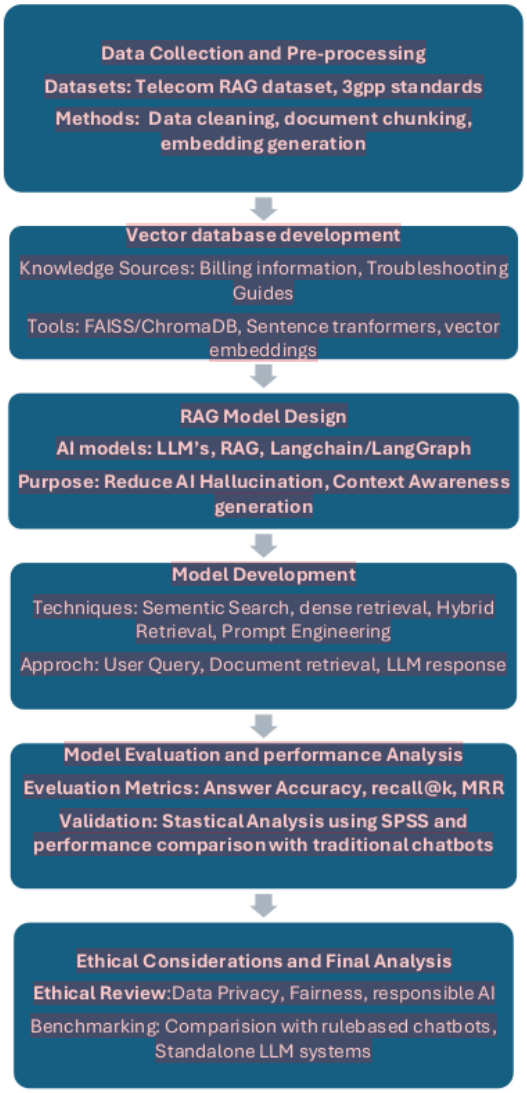


Figure 1: Framework Flowchart

8. Requirements and Resources

Software:

Microsoft Excel:

Microsoft Excel will be employed during the initial stages of the research for organising, inspecting, and preparing the collected datasets. It will assist in identifying missing or inconsistent values, performing preliminary data transformations, and generating descriptive summaries and visual representations. These activities will support data understanding before the datasets are processed within the proposed RAG framework.

IBM SPSS Statistics: IBM SPSS Statistics will be used to perform quantitative analysis and evaluate the effectiveness of the proposed framework. The software will support descriptive and inferential statistical techniques, including regression analysis, hypothesis testing, independent sample t-tests, ANOVA, and correlation analysis where appropriate. Statistical findings will be used to compare the proposed RAG-enabled customer support system with conventional chatbot approaches across evaluation measures such as response accuracy, retrieval efficiency, customer satisfaction, and overall system performance.

Python: Python will be used to implement the Retrieval-Augmented Generation framework. It will support data preprocessing, embedding generation, semantic retrieval, Large Language Model integration, evaluation, and visualization using libraries such as LangChain, LlamaIndex, Hugging Face Transformers, Pandas, NumPy, and Scikit-learn.

Hardware:

Computing Resources: The research will involve the computer with sufficient processing power to develop and evaluate the RAG framework. A workstation equipped with an Intel Core i7 or AMD Ryzen 7 processor, at least 16 GB RAM, SSD storage, and access to an NVIDIA GPU will provide adequate computational resources for embedding generation and model evaluation, also requires the cloud platforms like AWS(Sagemaker AI), Azure(AI foundary), Google cloud(Vertex AI) used.

Table 2: Software and hardware requirements for the study

Software	Hardware	Datasets
Microsoft Excel, IBM SPSS Statistics, Python, LangChain, LlamaIndex, Hugging Face Transformers, Pandas, NumPy,	Intel Core i7/AMD Ryzen 7 or Xeon processor, 16–32 GB RAM, SSD storage (minimum 256 GB), NVIDIA GPU (recommended),	TelecomRAG (Yilma et al., 2025), Telco-ORAG (Bornea et al., 2025), RAG Customer Support Dataset (Sukhwal & Analytics, 2024), Telecom FAQs, 3GPP standards,

Scikit-learn, FAISS (or ChromaDB), Matplotlib	AWS SageMaker, Azure AI Foundry, or Google Vertex AI	troubleshooting manuals, customer service documentation
Supports data preparation, statistical evaluation, implementation of the RAG pipeline, semantic retrieval, visualisation, and performance assessment	Provides the computational capability required for embedding generation, vector indexing, LLM inference, and experimental evaluation	Supplies domain-specific knowledge for retrieval, response generation, benchmarking, and evaluation of the proposed telecom customer support framework

9. Research Plan

Phase 1: Literature Review and Data Collection(2 Weeks)

The research will begin with a systematic examination of academic publications, industry reports, technical standards, and recent studies related to Generative Artificial Intelligence, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), semantic information retrieval, vector databases, and intelligent telecom customer support systems. This review aims to establish the current state of research, identify unresolved challenges, and determine opportunities for improving AI-powered customer support within the telecommunications industry.

Relevant datasets and domain-specific documents will then be collected to support the development of the proposed framework. These resources include TelecomRAG (Yilma et al., 2025), Telco-ORAG (Bornea et al., 2025), the RAG Customer Support Dataset (Sukhwal & Analytics, 2024), telecom Frequently Asked Questions (FAQs), troubleshooting guides, billing documentation, customer service manuals, and 3GPP technical standards. Together, these materials will provide the knowledge repository required for implementing the Retrieval-Augmented Generation system.

Phase 2: Data Preprocessing and Model Development(3 Weeks)

In this step, the data will be preprocessed and arranged in excel will be cleaned and prepared for use in the proposed system. This process will include removing duplicate records, correcting formatting issues, normalising the text, and dividing large documents into smaller sections to improve retrieval performance.After preprocessing, document embeddings will be generated and stored in a vector database such as FAISS or ChromaDB. The Retrieval-

Augmented Generation framework will then be developed by integrating the vector database with a Large Language Model

Phase 3: Model Evaluation and Statistical Analysis (3 weeks):

The completed RAG framework will be evaluated using quantitative performance measures, including response accuracy, precision, recall, F1-score, retrieval latency, and customer query resolution effectiveness. Statistical analysis will be performed using IBM SPSS Statistics to determine whether the proposed approach demonstrates measurable improvements over conventional rule-based customer support systems.

Descriptive statistics, correlation analysis, independent sample t-tests, ANOVA, and regression analysis will be applied where appropriate to examine relationships between evaluation variables and validate the research findings. The statistical outcomes will support conclusions regarding the reliability, effectiveness, and practical applicability of the proposed framework.

Phase 4: Final Review, Interpretability, and Analysis (3 Weeks)

The final phase will focus on interpreting the experimental findings and assessing the overall performance of the proposed RAG-based customer support framework. The generated responses will be analysed for contextual relevance, factual consistency, retrieval quality, and the occurrence of hallucinations. The strengths and limitations of the proposed solution will also be discussed in relation to existing telecom customer support methods.

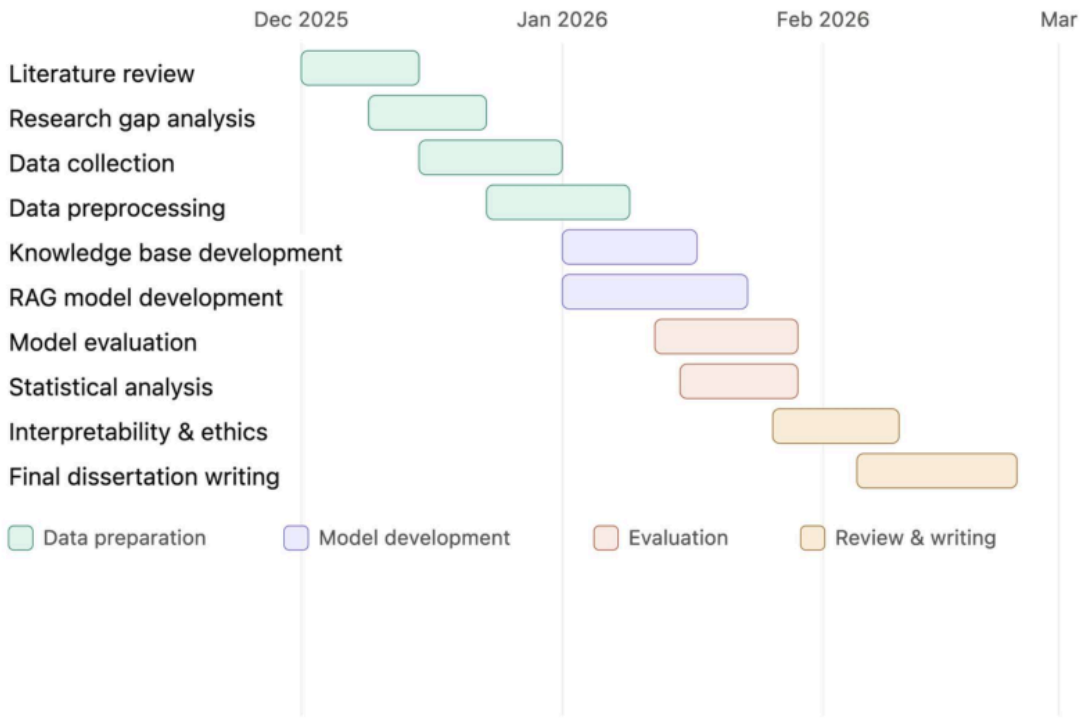


Figure 2: Proposed Research Plan

10. Risks and Contingency Plans

Several challenges may arise during the course of this research. These include limited availability of high-quality telecom datasets, incomplete or inconsistent technical documentation, retrieval errors, AI hallucinations, and computational resource limitations. Variations in telecom terminology across different datasets may also affect retrieval accuracy and the overall performance of the proposed system.

To minimise these risks, multiple publicly available telecom datasets will be combined to improve the quality and coverage of the knowledge base. Advanced retrieval techniques, including hybrid retrieval, reranking methods, and prompt optimisation, will be used to improve retrieval accuracy and reduce hallucinations. Cross-validation and repeated experimental testing will be carried out to ensure that the results are reliable and consistent. If local hardware resources become insufficient, cloud-based platforms such as Google Colab, AWS SageMaker, Azure AI Foundry, or Google Cloud Vertex AI will be used to complete the experiments successfully.

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APPENDIX B: ETHICAL COMPLIANCE STATEMENT

This research does not involve human participants, personal data collection, or interventions. All experiments were conducted exclusively on publicly available datasets, which were previously anonymized by their creators. As a result, no consent forms, confidentiality agreements, or participant information sheets were required.

Ethics Context for This Study

The study uses One open-source datasets:

https://www.kaggle.com/datasets/syncoraai/customer-support-conversations-Customer Support Conversations

Mentioned Dataset provides a fully synthetic collection of customer support interactions, generated using Syncora.ai's synthetic data generation engine. It mirrors realistic support conversations across e-commerce, banking, SaaS, and telecom domains, ensuring diversity, context depth, and privacy-safe realism.

Each conversation simulates multi-turn dialogues between a customer and a support agent, making it ideal for training chatbots, LLMs, and retrieval-augmented generation (RAG) systems.

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